UP-PCS PRELIMS 2026 Â· HIGH PROBABILITY QUESTION BANK

# Master Expected Questions — Part 2

High-Yield Questions 076 to 150 (Random Subject Distribution with UPPSC Traps & Ratta Points)

**QUESTION RANGE**
Questions 076 – 150

**PATTERN & STYLE**
UPPSC 2024â€"2025 Standard (Statements, A&R, Match Lists)

**STUDY FORMAT**
Question â†' Options â†' Answer â†' In-Depth Ratta & Trap Notes

**TARGET DATE**
December 6 Prelims Examination

# UPPCS Prelims 2026 Expected Questions — Part 2 (Q076 to Q150)

**Subject Distribution:** Randomized Full-Spectrum GS (History, Geography, Polity, Science, Environment, Economy, UP Special, Current Affairs)
**Exam Standard:** Multi-statement, Assertion-Reason, Match Lists & Direct Factual Traps as per UPPSC 2024–2025 Trends

## Question 76

**Subject:** Indian Polity | **Topic:** Fundamental Rights & Judicial Review (Article 13)

Consider the following statements regarding Article 13 of the Indian Constitution: 1. Under Article 13(2), any law made by Parliament or State Legislatures that abridges or takes away Fundamental Rights is void to the extent of the contravention. 2. The 'Doctrine of Eclipse' and 'Doctrine of Severability' have evolved primarily through judicial interpretation of Article 13. 3. The Supreme Court in the Kesavananda Bharati case held that a Constitutional Amendment Act under Article 368 is considered ordinary "law" within the meaning of Article 13(2).

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Statements 1 and 2 are correct: Article 13 provides the constitutional backbone for Judicial Review. Clause (2) mandates that the State shall not make any law which takes away or abridges Part III rights. The Supreme Court formulated the *Doctrine of Severability* (valid portions survive if invalid portions can be cleanly severed) and *Doctrine of Eclipse* (pre-constitutional laws inconsistent with FRs remain dormant/shadowed, not dead). - Statement 3 is incorrect: The Supreme Court in *Kesavananda Bharati (1973)* upheld the validity of clause (4) of Article 13 (inserted by 24th CAA 1971), affirming that a **Constitutional Amendment Act passed under Article 368 is NOT ordinary 'law' under Article 13(2)**. However, amendments can still be struck down if they violate the "Basic Structure". - **आयोग का Trap / Ratta Point:** Remember: The term "Judicial Review" is nowhere explicitly mentioned in the text of the Constitution, but Article 13, 32, 136, 226, and 227 vest this power explicitly in the higher judiciary!

## Question 77

**Subject:** World Geography | **Topic:** World Deserts & Adjacent Cold Currents

Match List-I (Desert of the World) with List-II (Continent / Region & Influencing Cold Current) and select the correct answer:

**List-I (Desert)** A. Atacama Desert
B. Namib Desert
C. Mojave Desert
D. Great Victoria Desert

**List-II (Region & Ocean Current)** 1. Southern Africa; Benguela Cold Current
2. South America (Chile/Peru); Humboldt (Peru) Cold Current
3. Australia; West Australian Cold Current
4. North America (USA); California Cold Current

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Atacama Desert (Northern Chile):** Driest non-polar desert on Earth; coastal fog (*Camanchaca*) exists due to the offshore Humboldt (Peru) cold current. - **Namib Desert (Namibia):** Ancient coastal desert formed by the desiccating effects of the cold Benguela Current. - **Mojave Desert (Southwestern USA):** Aridity reinforced by the cold California Current and rain-shadow of Sierra Nevada. - **Great Victoria Desert (Western & South Australia):** Aridity caused by subtropical high pressure and the cold West Australian Current. - **आयोग का Trap / Ratta Point:** Most tropical hot deserts are located on the **western margins of continents between 15° and 30° latitudes** due to two factors: Trade winds blowing offshore and Cold ocean currents running offshore!

## Question 78

**Subject:** Modern Indian History | **Topic:** Swadeshi Movement (1905–1908)

Consider the following statements regarding the Swadeshi and Boycott Movement: 1. The formal proclamation of the Swadeshi Movement was made on August 7, 1905, at the Calcutta Town Hall. 2. The 'Carlyle Circular' was issued by the British administration to prevent school and college students from participating in nationalist agitation. 3. The National Council of Education was set up in August 1906, with Aurobindo Ghosh as its first Principal.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - On August 7, 1905, the famous Boycott Resolution was passed at Calcutta Town Hall, inaugurating the Swadeshi Movement. - Chief Secretary Carlyle issued the repressive 'Carlyle Circular' threatening to withdraw grants, scholarships, and university affiliations from institutions whose students joined protests. In response, nationalists established the Anti-Circular Society (led by Sachindra Prasad Bose). - On August 15, 1906, the National Council of Education was inaugurated, and the Bengal National College was founded with Sri Aurobindo Ghosh as its principal. - **आयोग का Trap / Ratta Point:** On 16 October 1905 (the day Partition took effect), Rabindranath Tagore suggested observing the day as **Raksha Bandhan Day** to demonstrate Hindu-Muslim brotherhood, and composed *Amar Shonar Bangla*!

## Question 79

**Subject:** General Science (Biology) | **Topic:** Human Diseases, Pathogens & Diagnostic Tests

Match List-I (Disease) with List-II (Causative Organism & Diagnostic Test) and select the correct answer:

**List-I (Disease)** A. Tuberculosis
B. Typhoid Fever
C. Malaria
D. Kala-Azar (Visceral Leishmaniasis)

**List-II (Organism & Vector / Test)** 1. *Salmonella typhi* bacteria; Widal Test
2. *Mycobacterium tuberculosis* bacteria; Mantoux Tuberculin Skin Test
3. *Leishmania donovani* protozoan; transmitted by Sandfly (*Phlebotomus*)
4. *Plasmodium* protozoan; transmitted by female *Anopheles* mosquito

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Tuberculosis (TB):** Caused by *Mycobacterium tuberculosis*; airborne transmission; diagnosed via Mantoux test and sputum microscopy; vaccine is BCG (Bacillus Calmette-Guérin). - **Typhoid:** Caused by *Salmonella typhi*; transmitted via contaminated food/water; diagnosed via the **Widal Test**. - **Malaria:** Caused by protozoan parasite *Plasmodium* (falciparum, vivax, malariae, ovale); vector is female *Anopheles* mosquito; treated with Quinine (extracted from Cinchona tree bark) and Artemisinin. - **Kala-Azar (Black Fever):** Caused by protozoan *Leishmania donovani*; vector is the female **Sandfly** (*Phlebotomus*); targets bone marrow, spleen, and liver. - **आयोग का Trap / Ratta Point:** Vector matching trap: Sleeping Sickness = Tsetse fly; Dengue & Chikungunya = *Aedes aegypti* mosquito; Filariasis = *Culex* mosquito; Plague = Rat flea (*Xenopsylla cheopis*).

## Question 80

**Subject:** Uttar Pradesh Special | **Topic:** River Drainage of Uttar Pradesh

Consider the following statements regarding the rivers of Uttar Pradesh: 1. Gomti river originates from Fulhar Lake (Gomat Taal) near Madho Tanda in Pilibhit district of Uttar Pradesh. 2. The Betwa river joins the Yamuna river near Hamirpur in Uttar Pradesh. 3. Ken river joins the Yamuna river at Chilla near Banda in Uttar Pradesh.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Gomti:** An exceptional alluvial plain river; originates not from the Himalayas, but from Gomat Taal / Fulhar lake in **Pilibhit**; flows through Lucknow, Sultanpur, and Jaunpur before meeting the Ganga at Kaithi (Ghazipur/Varanasi border). - **Betwa (Vetravati):** Originates in Raisen (MP); enters UP in Lalitpur and Jhansi; joins Yamuna near **Hamirpur**. - **Ken (Karnavati):** Originates in Kaimur hills (MP); enters Banda district of UP and merges with Yamuna at **Chilla** (Banda). - **आयोग का Trap / Ratta Point:** The Ken-Betwa River Interlinking Project transfers surplus water from Ken basin to Betwa basin to irrigate the drought-prone Bundelkhand region (submerging part of Panna Tiger Reserve).

## Question 81

**Subject:** Environment & Ecology | **Topic:** Biodiversity Hotspots in India

The concept of 'Biodiversity Hotspots' was first introduced by Norman Myers in 1988. How many biodiversity hotspots cover parts of India?

**Options:** A. 2
B. 3
C. 4
D. 5

**Answer:** C

**Explanation:** - **Core Fact & Solution:** - Globally, Conservation International recognizes 36 Biodiversity Hotspots. - **4 Hotspots cover parts of India:** 1. **The Western Ghats:** High degree of endemism (Nilgiri Tahr, Lion-tailed Macaque, Malabar Civet). 2. **The Himalayas:** Entire Indian Himalayan region (spanning J&K, Ladakh, Himachal, Uttarakhand, Sikkim, Arunachal Pradesh). 3. **Indo-Burma:** Covers Northeastern India (south of the Brahmaputra river) and Andaman Islands. 4. **Sundaland:** Covers the Nicobar group of Islands (extending to Malaysia, Indonesia, Brunei). - **आयोग का Trap / Ratta Point:** Qualifying criteria for a Hotspot: Must contain at least **1,500 species of vascular plants as endemics** (> 0.5% of world total) AND must have lost at least **70% of its original primary habitat**!

## Question 82

**Subject:** Indian Economy | **Topic:** Budget Deficits & Fiscal Indicators

Match List-I (Type of Deficit) with List-II (Formula / Definition) and select the correct answer:

**List-I (Deficit)** A. Fiscal Deficit
B. Revenue Deficit
C. Primary Deficit
D. Effective Revenue Deficit

**List-II (Formula / Definition)** 1. Fiscal Deficit minus Interest Payments
2. Revenue Deficit minus Grants for Creation of Capital Assets
3. Revenue Expenditure minus Revenue Receipts
4. Total Expenditure minus Total Receipts excluding borrowings

**Options:** A. A-4, B-3, C-1, D-2
B. A-3, B-4, C-1, D-2
C. A-4, B-1, C-3, D-2
D. A-2, B-3, C-1, D-4

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Fiscal Deficit:** Total Expenditure - (Revenue Receipts + Non-debt Capital Receipts). Represents the total borrowing requirements of the government from all sources. - **Revenue Deficit:** Revenue Expenditure - Revenue Receipts. Indicates that the government is dissaving and using borrowings to finance routine current consumption. - **Primary Deficit:** Fiscal Deficit - Interest Payments. Indicates the borrowing requirements of the government excluding the interest liability on past accumulated debt. - **Effective Revenue Deficit:** Introduced in Union Budget 2011–12 (amended FRBM Act); excludes grants given to States/local bodies for the creation of capital assets from Revenue Deficit. - **आयोग का Trap / Ratta Point:** If Primary Deficit is zero, it means: Fiscal Deficit = Interest Payments (i.e., the entire borrowing is being consumed just to pay interest on past loans!).

## Question 83

**Subject:** Current Affairs | **Topic:** Cinema & Cultural Awards (Dadasaheb Phalke 2024)

Who was conferred with the prestigious Dadasaheb Phalke Award—India's highest recognition in the field of cinema—for the year 2022 at the 70th National Film Awards ceremony held in October 2024?

**Options:** A. Mithun Chakraborty
B. Waheeda Rehman
C. Asha Parekh
D. Rajinikanth

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Veteran actor and cultural icon **Mithun Chakraborty** was conferred the Dadasaheb Phalke Award at the 70th National Film Awards in October 2024 for his distinguished contribution to Indian cinema (debuted in Mrinal Sen's *Mrigayaa* in 1976, winning National Award for Best Actor). - Previous recipients: Waheeda Rehman (for 2021), Asha Parekh (for 2020), Rajinikanth (for 2019), Amitabh Bachchan (for 2018). - **आयोग का Trap / Ratta Point:** At the 70th National Film Awards, Best Feature Film was awarded to *Aattam* (Malayalam), and Rishab Shetty won Best Actor for *Kantara*.

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## Question 84

**Subject:** Ancient Indian History | **Topic:** Sangam Literature & Ancient Ports

Consider the following statements regarding the Sangam Age: 1. *Tolkappiyam*, authored by Tolkappiyar, is the oldest extant Tamil grammatical work. 2. The epic *Silappadikaram* was composed by Ilango Adigal, centering around the tragic story of Kovalan, Kannagi, and the Pandya king of Madurai. 3. Muziris was a celebrated port city of the Chola kingdom located on the Coromandel coast.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Statements 1 and 2 are correct: *Tolkappiyam* is a masterpiece of grammar, linguistics, and poetics dating to the Second Sangam. *Silappadikaram* (The Jeweled Anklet) is an epic written by Jain prince Ilango Adigal about Kannagi's vengeance on Madurai. - Statement 3 is incorrect: **Muziris** (modern Kodungallur / Pattanam in Kerala) was the premier ancient port of the **Chera kingdom on the Malabar (western) coast**, famous for Roman pepper trade. The chief port of the Cholas was **Puhar (Kaveripattinam)** on the eastern Coromandel coast! - **आयोग का Trap / Ratta Point:** Sangam kingdoms and their capitals & ports: - **Cheras:** Capital = Vanji / Karur; Port = Muziris & Tondi; Emblem = Bow. - **Cholas:** Capital = Uraiyur / Puhar; Port = Kaveripattinam (Puhar); Emblem = Tiger. - **Pandyas:** Capital = Madurai; Port = Korkai; Emblem = Carp (Fish).

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## Question 85

**Subject:** Geography of India | **Topic:** Godavari River System & Tributaries

Which of the following is a LEFT-BANK tributary of the Godavari River?

**Options:** A. Manjira
B. Penganga
C. Pranhita
D. Both B and C

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - The Godavari (known as *Dakshin Ganga* or *Vriddha Ganga*) originates from Trimbakeshwar in the Nashik district of Maharashtra; length 1,465 km. - **Left-Bank Tributaries:** Dharna, Penganga, Wainganga, Wardha, **Pranhita** (formed by the confluence of Wardha and Wainganga), Pench, Kanhan, Sabari, and Indravati. - **Right-Bank Tributaries:** Pravara, Mula, and **Manjira** (Manjira is the only major right-bank tributary, originating in Balaghat range). - **आयोग का Trap / Ratta Point:** Manjira is a RIGHT-bank tributary; Indravati and Pranhita are LEFT-bank tributaries!

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## Question 86

**Subject:** Indian Polity | **Topic:** Fundamental Duties (Article 51A)

Consider the following statements regarding the Fundamental Duties in the Indian Constitution: 1. Fundamental Duties were incorporated into Part IV-A of the Constitution upon the recommendation of the Swaran Singh Committee. 2. The 86th Constitutional Amendment Act, 2002 added the 11th Fundamental Duty regarding education of children between 6 and 14 years. 3. Fundamental Duties are directly enforceable by writs issued under Article 32.

Which of the statements given above is/are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 only
D. 1, 2 and 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Statements 1 and 2 are correct: The Swaran Singh Committee (1976) recommended 8 duties; the 42nd CAA 1976 inserted Article 51A with 10 duties borrowed from the USSR constitution. The 86th CAA 2002 added the 11th duty [Article 51A(k)]: duty of a parent or guardian to provide educational opportunities to their child between ages 6 and 14. - Statement 3 is incorrect: Fundamental Duties, like DPSPs, are **non-justiciable and non-enforceable by judicial writs**. However, Parliament can enforce them by enacting specific substantive laws (e.g. Prevention of Insults to National Honour Act 1971, Wildlife Protection Act 1972). - **आयोग का Trap / Ratta Point:** Swaran Singh Committee had recommended a "Duty to Pay Taxes" and penalties for non-compliance, but the Indira Gandhi government REJECTED the duty to pay taxes and penalties!

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## Question 87

**Subject:** Medieval Indian History | **Topic:** Administration of Sher Shah Suri (1540–1545)

Consider the following statements regarding Sher Shah Suri: 1. He introduced the silver coin called 'Rupiya' (178 grains) and copper coin called 'Dam' (380 grains), which formed the basis of the currency system up to the British period. 2. He revived the grand highway running from Sonargaon (Bengal) to Peshawar, known as the *Sadak-i-Azam* (later called Grand Trunk Road). 3. His monumental octagonal tomb situated in the middle of an artificial lake is located at Sasaram in Bihar.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Currency Reform:** Sher Shah abolished all debased mixed-metal coins and introduced standard coins: silver *Rupiya* (178 grains of pure silver) and copper *Dam* (ratio was 1 Rupiya = 64 Dams). - **Communications:** Reconstructed the imperial highway connecting Sonargaon (East Bengal) through Agra and Delhi to Attock/Peshawar (*Sadak-i-Azam*), constructing 1,700 *Sarais* (inns) with separate accommodation for Hindus and Muslims. - **Architecture:** Built the majestic red sandstone octagonal tomb at **Sasaram** (Rohtas district, Bihar) standing in an artificial lake; also built the *Purana Qila* (Qila-i-Kuhna Mosque) in Delhi. - **आयोग का Trap / Ratta Point:** Sher Shah died in May 1545 during the siege of **Kalinjar Fort** (Banda, Bundelkhand, UP) due to an accidental gunpowder explosion of a rocket named *Uqqa*!

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## Question 88

**Subject:** Uttar Pradesh Special | **Topic:** Cultural Folk Music & Occasions

Match List-I (Folk Music Tradition of UP) with List-II (Core Theme / Occasion) and select the correct answer:

**List-I (Tradition)** A. Birha
B. Kajari
C. Alha
D. Rasiya

**List-II (Theme / Occasion)** 1. Heroic tales of medieval warriors and martial chivalry
2. Songs celebrating Lord Krishna and Radha, sung during Holi in Braj
3. Yearning for the beloved, sung during monsoon season by women
4. Narrative ballads depicting social events and heroism, sung in Purvanchal

**Options:** A. A-4, B-3, C-1, D-2
B. A-3, B-4, C-1, D-2
C. A-4, B-1, C-3, D-2
D. A-2, B-3, C-1, D-4

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Birha:** Bhojpuri/Purvanchal narrative singing; recounts folklore, social satire, and heroism. - **Kajari:** Sung in Mirzapur/Varanasi during monsoon/Sawani teej; expresses separation, longing, and joy of rainy season. - **Alha:** Sung in Bundelkhand; tells martial ballads of Alha and Udal fighting against Prithviraj Chauhan. - **Rasiya:** Sung in Braj; filled with devotion, romance, and playful lyrics of Radha-Krishna during Holi. - **आयोग का Trap / Ratta Point:** Sohar is sung on the birth of a child (especially in Purvanchal and Awadh). Hori/Dhamar is sung during Holi.

## Question 89

**Subject:** General Science (Chemistry) | **Topic:** Polymers, Plastics & Synthetic Fibres

Match List-I (Polymer / Substance) with List-II (Monomer / Unique Application) and select the correct answer:

**List-I (Polymer)** A. Teflon
B. Natural Rubber
C. Bakelite
D. Polyvinyl Chloride (PVC)

**List-II (Monomer / Application)** 1. Isoprene (2-methyl-1,3-butadiene)
2. Polytetrafluoroethylene; non-stick coating for cookware
3. Thermosetting polymer of Phenol and Formaldehyde; electrical switches
4. Vinyl Chloride; pipes, insulation, and raincoats

**Options:** A. A-2, B-1, C-3, D-4
B. A-1, B-2, C-3, D-4
C. A-2, B-3, C-1, D-4
D. A-4, B-1, C-3, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Teflon (PTFE):** Formed by polymerization of tetrafluoroethylene ($F_2C=CF_2$); high thermal stability and chemical inertness; used for non-stick pans and gasket seals. - **Natural Rubber:** Addition polymer of **Isoprene** (cis-1,4-polyisoprene). Vulcanization involves heating natural rubber with Sulphur to introduce cross-links and elasticity. - **Bakelite:** Oldest synthetic **thermosetting plastic**; condensation polymer of Phenol and Formaldehyde; infusible once molded; used for electrical switches, plugs, and cooker handles. - **PVC:** Polymer of vinyl chloride ($CH_2=CHCl$); widely used for water pipes and cable sheaths. - **आयोग का Trap / Ratta Point:** Distinction: Bakelite is a *thermosetting* plastic (cannot be remolded on heating); PVC and Polythene are *thermoplastics* (can be melted and reshaped repeatedly).

## Question 90

**Subject:** Environment & Ecology | **Topic:** Air Quality Index (AQI) Parameters

The National Air Quality Index (AQI) launched under the Swachh Bharat Abhiyan in India measures the concentration of how many atmospheric pollutants?

**Options:** A. 6
B. 8
C. 10
D. 12

**Answer:** B

**Explanation:** - **Core Fact & Solution:** - The National AQI monitors **8 criteria pollutants**: 1. Particulate Matter 10 ($PM_{10}$) 2. Particulate Matter 2.5 ($PM_{2.5}$) 3. Nitrogen Dioxide ($NO_2$) 4. Sulphur Dioxide ($SO_2$) 5. Carbon Monoxide ($CO$) 6. Ozone ($O_3$) 7. Ammonia ($NH_3$) 8. Lead ($Pb$) - AQI has 6 color-coded categories: Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very Poor (301–400), and Severe (401–500). - **आयोग का Trap / Ratta Point:** Carbon Dioxide ($CO_2$) is NOT one of the 8 pollutants measured by AQI! (UPPSC loves testing if $CO_2$ is in AQI — it is an atmospheric greenhouse gas, but not an AQI criteria pollutant).

## Question 91

**Subject:** Indian Economy | **Topic:** Priority Sector Lending (PSL) Targets

Consider the following statements regarding Priority Sector Lending (PSL) norms mandated by the Reserve Bank of India: 1. Domestic scheduled commercial banks are required to allocate 40% of their Adjusted Net Bank Credit (ANBC) to priority sectors. 2. The sub-target for Agriculture lending is 18% of ANBC. 3. The sub-target for Micro Enterprises is 7.5% of ANBC.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - Overall PSL target for domestic commercial banks and foreign banks with 20+ branches: **40% of ANBC** (or CEOBE). - **Agriculture Sub-target:** 18% of ANBC (within which 10% is earmarked for Small and Marginal Farmers - SMFs). - **Micro Enterprises Sub-target:** 7.5% of ANBC. - **Weaker Sections Sub-target:** 12% of ANBC. - Regional Rural Banks (RRBs) and Small Finance Banks (SFBs) have a higher overall target of **75% of ANBC**. - **आयोग का Trap / Ratta Point:** If banks fall short of their PSL targets, the shortfall must be deposited into the **Rural Infrastructure Development Fund (RIDF)** maintained by NABARD!

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## Question 92

**Subject:** Current Affairs | **Topic:** Space & Defence (Aditya-L1 Mission)

Consider the following statements regarding ISRO's Aditya-L1 Mission: 1. It is India's first dedicated solar observatory in space. 2. It was successfully inserted into a halo orbit around the Sun-Earth Lagrange Point 1 (L1), located approximately 1.5 million kilometres from Earth. 3. The Visible Emission Line Coronagraph (VELC) is the primary payload designed to observe the solar corona.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - Aditya-L1 was launched on September 2, 2023, aboard PSLV-C57 from Satish Dhawan Space Centre, Sriharikota. - On January 6, 2024, it achieved final halo-orbit insertion around **Lagrange Point 1 (L1)**, ~1.5 million km from Earth (~1% of Earth-Sun distance). - Carries 7 scientific payloads: 4 remote sensing payloads including **VELC** (developed by Indian Institute of Astrophysics, Bengaluru) and **SUIT** (Solar Ultraviolet Imaging Telescope), plus 3 in-situ payloads (ASPEX, PAPA, and Magnetometer). - **आयोग का Trap / Ratta Point:** A satellite in halo orbit around L1 enjoys the major advantage of continuously viewing the Sun without any occultation or eclipses!

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## Question 93

**Subject:** Modern Indian History | **Topic:** Quit India Movement (1942)

Consider the following statements regarding the Quit India Movement: 1. The resolution was ratified at the Gowalia Tank Maidan session of the All India Congress Committee in Bombay on August 8, 1942. 2. Under 'Operation Zero Hour', all top leaders of Congress were arrested in early morning raids on August 9, 1942. 3. During the movement, Chittu Pandey established a parallel national government at Ballia in Uttar Pradesh.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - Gandhi gave the historic call "Do or Die" (*Karo ya Maro*) at Gowalia Tank (August Kranti Maidan), Bombay. - British government launched **Operation Zero Hour** at dawn on August 9, arresting

Mahatma Gandhi (confined at Aga Khan Palace, Pune) and CWC members (confined at Ahmednagar Fort). - **Parallel Governments (*Prati Sarkar*):** - **Ballia (UP):** Established for a week under Chittu Pandey ("Sher-e-Ballia"). - **Tamluk (Midnapore, Bengal):** *Jatiya Sarkar* (featuring Matangini Hazra). - **Satara (Maharashtra):** Longest lasting *Prati Sarkar* led by Nana Patil and Y.B. Chavan. - **आयोग का Trap / Ratta Point:** Secret underground radio during Quit India Movement was operated by **Usha Mehta** and Babubhai Prasad from Bombay!

## Question 94

**Subject:** World Geography | **Topic:** Major Canals and Waterways

Match List-I (Canal) with List-II (Bodies of Water Connected) and select the correct answer:

**List-I (Canal)** A. Suez Canal
B. Panama Canal
C. Kiel Canal
D. Soo Canal

**List-II (Bodies Connected)** 1. Atlantic Ocean and Pacific Ocean
2. Mediterranean Sea and Red Sea
3. Lake Superior and Lake Huron
4. North Sea and Baltic Sea

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Suez Canal (Egypt, opened 1869):** Engineered by Ferdinand de Lesseps; connects the Mediterranean Sea (Port Said) with the Red Sea (Port Tewfik/Suez); cuts sea route between Europe and Asia by ~7,000 km. - **Panama Canal (opened 1914):** Cuts through Isthmus of Panama; connects the Atlantic Ocean (Caribbean Sea) with the Pacific Ocean utilizing a lock system. - **Kiel Canal (Germany):** Connects the North Sea with the Baltic Sea; busiest artificial waterway in the world. - **Soo Canal (USA/Canada):** Connects Lake Superior with Lake Huron bypass around St. Mary's Falls. - **आयोग का Trap / Ratta Point:** Suez Canal has NO locks (sea-level canal throughout), whereas Panama Canal uses an intricate 3-tiered lock system (Gatun, Pedro Miguel, Miraflores)!

## Question 95

**Subject:** Indian Polity | **Topic:** Parliamentary Financial Committees

Match List-I (Parliamentary Committee) with List-II (Composition & Salient Feature) and select the correct answer:

**List-I (Committee)** A. Public Accounts Committee (PAC)
B. Estimates Committee
C. Committee on Public Undertakings

**List-II (Composition & Feature)** 1. 30 members, ALL from Lok Sabha only; largest committee of Parliament
2. 22 members (15 Lok Sabha + 7 Rajya Sabha); examines reports of CAG
3. 22 members (15 Lok Sabha + 7 Rajya Sabha); examines PSUs and Krishna Menon Committee recommendations

**Options:** A. A-2, B-1, C-3
B. A-1, B-2, C-3
C. A-2, B-3, C-1
D. A-3, B-1, C-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **PAC:** Oldest financial committee (set up in 1921 under GOI Act 1919); 22 members (15 LS, 7 RS); by convention since 1967, its chairman is from the Opposition; acts as watchdog over government spending based on CAG audit reports. - **Estimates Committee:** Constituted in 1950 on recommendations of Finance Minister John Mathai; has **30 members, ALL drawn from Lok Sabha**; Rajya Sabha has NO representation; known as "Continuous Economy Committee". - **Committee on Public Undertakings:** Created in 1964 on recommendation of Krishna Menon Committee; 22 members (15 LS, 7 RS). - **आयोग का Trap / Ratta Point:** A **Minister CANNOT be elected** as a member of ANY of these three financial committees!

## Question 96

**Subject:** Ancient Indian History | **Topic:** Mauryan Administration Officers (Arthashastra)

Match List-I (Mauryan Officer) with List-II (Department / Portfolio) and select the correct answer:

**List-I (Officer)** A. Samaharta
B. Sannidhata
C. Sitadhyaksha
D. Samsthadhyaksha

**List-II (Portfolio)** 1. Superintendent of Crown Agricultural Lands
2. Chief Collector of Revenue and Assessor
3. Superintendent of Markets / Trade routes
4. Chief Custodian of the State Treasury and Storehouse

**Options:** A. A-2, B-4, C-1, D-3
B. A-4, B-2, C-1, D-3
C. A-2, B-1, C-4, D-3
D. A-3, B-4, C-1, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Samaharta:** Chief collector of revenue; responsible for assessing and collecting revenues across the entire realm. - **Sannidhata:** Chief treasurer and in-charge of royal storehouses (*Koshadhyaksha*). - **Sitadhyaksha:** Superintendent of *Sita* (crown agricultural lands cultivated by slaves, prisoners, and state tenants). - **Samsthadhyaksha:** Superintendent of markets, preventing fraud and regulating commercial trade. - **आयोग का Trap / Ratta Point:** Samaharta = Collector General; Sannidhata = Treasurer. Do not mix them up (UPPSC frequently swaps Samaharta and Sannidhata).

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## Question 97

**Subject:** General Science (Physics) | **Topic:** Sound Waves & Frequencies

Match List-I (Wave Category) with List-II (Frequency Range / Characteristic) and select the correct answer:

**List-I (Category)** A. Infrasonic Waves
B. Audible Sound Waves
C. Ultrasonic Waves
D. Mach Number

**List-II (Range / Characteristic)** 1. Frequencies above 20,000 Hz; used in SONAR and Echocardiography
2. Frequencies below 20 Hz; produced by earthquakes, volcanoes, and whales
3. Frequencies between 20 Hz and 20,000 Hz
4. Ratio of the speed of a moving body to the speed of sound in the same medium

**Options:** A. A-2, B-3, C-1, D-4
B. A-3, B-2, C-1, D-4
C. A-2, B-1, C-3, D-4
D. A-1, B-2, C-3, D-4

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Infrasonic (<20 Hz):** Inaudible to humans; elephants, rhinoceroses, and earthquakes generate infrasound. - **Audible (20 Hz – 20,000 Hz):** Normal human hearing range. - **Ultrasonic (>20,000 Hz):** Produced by bats and dolphins for echolocation; applied in SONAR (Sound Navigation and Ranging) to map ocean depth and ultrasonography (USG). - **Mach Number:** $ ext{Mach} = rac{v_{ ext{object}}}{v_{ ext{sound}}}$. Mach > 1 is Supersonic; Mach > 5 is Hypersonic. - **आयोग का Trap / Ratta Point:** Sound CANNOT travel in vacuum (requires a material medium). Speed of sound is fastest in Solids (Steel $pprox 5,960 ext{ m/s}$), slower in Liquids (Water $pprox 1,480 ext{ m/s}$), and slowest in Gases (Air $pprox 343 ext{ m/s}$).

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## Question 98

**Subject:** Uttar Pradesh Special | **Topic:** UP Census 2011 Demographic Extremes

Consider the following pairs regarding Uttar Pradesh Census 2011: 1. District with highest Sex Ratio: Jaunpur (1,024 females per 1,000 males) 2. District with lowest Sex Ratio: Gautam Buddha Nagar (851 females per 1,000 males) 3. District with highest Population Density: Ghaziabad (3,971 persons/sq km) 4. District with lowest Population Density: Lalitpur (242 persons/sq km)

How many of the above pairs are correctly matched?

**Options:** A. Only two pairs
B. Only three pairs
C. All four pairs
D. Only one pair

**Answer:** C

**Explanation:** - **Core Fact & Solution:** - All four pairs are perfectly matched: - **Sex Ratio (State average = 912):** - *Highest:* **Jaunpur (1024)**, Azamgarh (1019), Deoria (1017). - *Lowest:* **Gautam Buddha Nagar (851)**, Hamirpur & Kanpur Nagar (861). - **Population Density (State average = 829 per sq km):** - *Highest:* **Ghaziabad (3971)**, Varanasi (2395), Lucknow (1816). - *Lowest:* **Lalitpur (242)**, Sonbhadra (270), Hamirpur (275). - **आयोग का Trap / Ratta Point:** Gautam Buddha Nagar has the HIGHEST literacy in UP, but the LOWEST sex ratio in UP!

## Question 99

**Subject:** Environment & Ecology | **Topic:** Project Tiger & National Tiger Conservation Authority

Consider the following statements regarding Project Tiger in India: 1. Project Tiger was launched on April 1, 1973, from Jim Corbett National Park. 2. The National Tiger Conservation Authority (NTCA) is a statutory body constituted under the Wildlife (Protection) Act, 1972. 3. India holds more than 70% of the world's wild tiger population.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Launch:** April 1, 1973, under PM Indira Gandhi from Corbett National Park (Uttarakhand) with Kailash Sankhala as its first director ("Tiger Man of India"). - **NTCA:** Established following the 2006 amendment to the Wildlife (Protection) Act 1972 (chaired by the Union Minister of Environment, Forest and Climate Change). - **Population:** As per the 5th cycle of the All India Tiger Estimation (2022/2023), India's minimum tiger count reached **3,167** (estimated upper bound 3,925), accounting for over 75% of global wild tigers. - **आयोग का Trap / Ratta Point:** State with maximum tigers = **Madhya Pradesh (785 tigers - "Tiger State")**, followed by Karnataka (563) and Uttarakhand (560).

## Question 100

**Subject:** Indian Economy | **Topic:** Goods and Services Tax (GST) & GST Council

Consider the following statements regarding the Goods and Services Tax (GST): 1. GST was introduced in India by the 101st Constitutional Amendment Act, 2016. 2. The GST Council is established under Article 279A of the Constitution. 3. In the GST Council, the Central Government has a weightage of one-half of the total votes cast, while all State Governments combined have one-half.

Which of the statements given above is/are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 only
D. 1, 2 and 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Statements 1 and 2 are correct: GST came into effect on July 1, 2017 (101st CAA 2016). Article 279A empowered the President to constitute the GST Council within 60 days (chaired by the Union Finance Minister). - Statement 3 is incorrect: In the voting mechanism of the GST Council: - The vote of the **Central Government has a weightage of one-third ($↑rac{1}{3}$)** of total votes cast. - The votes of all **State Governments combined have a weightage of two-thirds ($↑rac{2}{3}$)** of total votes cast. - Every decision requires a majority of not less than **three-fourths ($↑rac{3}{4}$ / 75%)** of the weighted votes of members present and voting. - **आयोग का Trap / Ratta Point:** Voting weights trap: Central Govt = $↑rac{1}{3}$, State Govts = $↑rac{2}{3}$, Approval threshold = $↑rac{3}{4}$!

## Question 101

**Subject:** Current Affairs | **Topic:** Global Rankings & Indexes 2024

Match List-I (Global Report / Index) with List-II (Publishing Organization) and select the correct answer:

**List-I (Report / Index)** A. Global Innovation Index (GII)
B. World Press Freedom Index
C. Global Gender Gap Report
D. Logistics Performance Index (LPI)

**List-II (Publishing Body)** 1. Reporters Without Borders (RSF)
2. World Intellectual Property Organization (WIPO)
3. World Bank
4. World Economic Forum (WEF)

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Global Innovation Index:** Published annually by **WIPO**; India climbed to 39th rank in GII 2024 (Switzerland ranked 1st). - **World Press Freedom Index:** Published by Paris-based **Reporters Sans Frontières (RSF)**; Norway ranked 1st. - **Global Gender Gap Report:** Published by **World Economic Forum (WEF)** across 4 dimensions: Economic Participation, Educational Attainment, Health & Survival, Political Empowerment. - **Logistics Performance Index:** Published by **World Bank**; evaluates customs, infrastructure, international shipments, logistics competence, tracking & tracing, and timeliness. - **आयोग का Trap / Ratta Point:** World Development Report is published by World Bank; World Economic Outlook is published by IMF; Global Financial Stability Report is published by IMF!

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## Question 102

**Subject:** Medieval Indian History | **Topic:** Religious Policy of Mughal Emperor Akbar

Arrange the following measures undertaken by Akbar in chronological order: 1. Abolition of Jizya tax 2. Establishment of the Ibadat Khana at Fatehpur Sikri 3. Promulgation of the Mahzarnama (Infallibility Decree) 4. Promulgation of the Tauhid-i-Ilahi (Din-i-Ilahi)

Select the correct answer from the codes given below:

**Options:** A. 1 — 2 — 3 — 4
B. 1 — 3 — 2 — 4
C. 2 — 1 — 3 — 4
D. 1 — 2 — 4 — 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** 1. **Abolition of Jizya (1564):** Abolished the discriminatory tax on non-Muslim subjects (Akbar had already abolished the pilgrim tax in 1563). 2. **Ibadat Khana (1575):** Hall of Prayer built at Fatehpur Sikri; initially open only to Sunni scholars, opened to all religions (Hindus, Jains, Parsis, Christians) in 1578. 3. **Mahzarnama (September 1579):** Drafted by Sheikh Mubarak (father of Abul Fazl and Faizi); declared Akbar the *Imam-i-Adil* (supreme arbiter on Islamic law in case of conflicting interpretations). 4. **Tauhid-i-Ilahi / Din-i-Ilahi (1582):** Socio-religious ethical order based on universal peace (*Sulh-i-Kul*); Birbal was the only Hindu noble who accepted it. - **आयोग का Trap / Ratta Point:** Chronology mnemonic: **Pilgrim tax (1563) -> Jizya (1564) -> Ibadat Khana (1575) -> Mahzar (1579) -> Din-i-Ilahi (1582) -> Ilahi Samvat (1584)**.

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## Question 103

**Subject:** Indian Polity | **Topic:** President of India (Electoral College & Impeachment)

Consider the following statements regarding the President of India: 1. The Electoral College for electing the President (Article 54) includes the elected members of both Houses of Parliament and elected members of the Legislative Assemblies of States and the UTs of Delhi, Puducherry, and Jammu & Kashmir. 2. Nominated members of Parliament do not participate in the presidential election, but they DO participate in the impeachment of the President under Article 61. 3. Elected members of State Legislative Assemblies participate in the presidential election, but they DO NOT participate in the impeachment of the President.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Election Electoral College (Article 54):** Only ELECTED members of LS, RS, and State Legislative Assemblies (+ Delhi & Puducherry added by 70th CAA 1992). Nominated members and Legislative Councils (MLCs) do NOT participate. - **Impeachment (Article 61):** Quasi-judicial procedure in Parliament. Handled solely by Parliament: ALL members of LS and RS (both ELECTED and NOMINATED) vote! - Members of State Legislative Assemblies have NO role in the impeachment process. - **आयोग का Trap / Ratta Point:** Contrast with Vice-President election: Electoral college for VP consists of ALL members of Parliament (both elected and nominated), and State Assemblies have zero role!

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## Question 104

**Subject:** Geography of India | **Topic:** Soils of India (ICAR Classification)

Match List-I (Soil Type) with List-II (Distinct Feature / Mineral Content) and select the correct answer:

**List-I (Soil)** A. Alluvial Soil
B. Black Soil (Regur)
C. Laterite Soil
D. Peaty / Organic Soil

**List-II (Feature)** 1. High clay content with montmorillonite; self-ploughing capacity; ideal for cotton
2. Rich in humus and organic matter; highly acidic; found in submerged coastal areas (Kuttanad)
3. Formed by intense leaching in tropical regions with high rainfall; rich in iron and aluminium oxides
4. Divided into Khadar and Bhangar; rich in potash, deficient in nitrogen and phosphorus

**Options:** A. A-4, B-1, C-3, D-2
B. A-1, B-4, C-3, D-2
C. A-4, B-3, C-1, D-2
D. A-2, B-1, C-4, D-3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Alluvial Soil:** Largest soil group (~40% of India's area); transported by rivers; Khadar is new fertile fine silt; Bhangar is older alluvium with calcareous concretions (*Kankar*). - **Black Soil (Regur):** Formed by weathering of Deccan basaltic trap; high water-retention capacity; swells when wet and shrinks when dry developing deep cracks ("self-ploughing"). - **Laterite Soil:** Formed by intense tropical leaching (washing away of silica); leaves behind hydrated oxides of iron and aluminium; acidic, poor fertility, suitable for cashew, tea, coffee with fertilizers. - **Peaty Soil:** Heavy, black, highly acidic soils with 40–50% organic matter (found in Kottayam and Alappuzha in Kerala, Sundarbans). - **आयोग का Trap / Ratta Point:** Indian soils are almost universally deficient in **Nitrogen, Phosphorus, and Humus** (NPK ratio recommended is 4:2:1).

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## Question 105

**Subject:** Modern Indian History | **Topic:** Revolutionary Societies & Conspiracy Cases

Match List-I (Revolutionary Society) with List-II (Founder / Key Leader) and select the correct answer:

**List-I (Society)** A. Abhinav Bharat
B. Anushilan Samiti (Calcutta)
C. Ghadar Party
D. Hindustan Socialist Republican Association (HSRA)

**List-II (Founder / Leader)** 1. Pramathanath Mitra & Barindra Kumar Ghosh
2. V.D. Savarkar (Vinayak Damodar Savarkar)
3. Chandrashekhar Azad & Bhagat Singh
4. Lala Har Dayal & Sohan Singh Bhakna

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Abhinav Bharat (1904, Nasik):** Secret society formed by Vinayak Damodar Savarkar and Ganesh Savarkar (evolved from *Mitra Mela* 1899). - **Anushilan Samiti (1902, Bengal):** Founded by Pramathanath Mitra (P. Mitra), Barindra Kumar Ghosh, and Jatindranath Banerjee. - **Ghadar Party (1913, San Francisco, USA):** Founded by Lala Har Dayal, Sohan Singh Bhakna (President), and Kartar Singh Sarabha; weekly newspaper *The Ghadar*. - **HSRA (September 1928, Feroz Shah Kotla, Delhi):** Reorganized from HRA by Chandrashekhar Azad, Bhagat Singh, Sukhdev, adding "Socialist" to the title. - **आयोग का Trap / Ratta Point:** HRA was founded in Kanpur in 1924 by Ram Prasad Bismil, Sachindra Nath Sanyal, and Jogesh Chandra Chatterjee. It was transformed into HSRA in Delhi in 1928!

## Question 106

**Subject:** General Science (Biology) | **Topic:** Digestive System & Enzymes

Match List-I (Digestive Enzyme) with List-II (Secretory Site & Substrate Digested) and select the correct answer:

**List-I (Enzyme)** A. Pepsin
B. Ptyalin (Salivary Amylase)
C. Trypsin
D. Lipase

**List-II (Site & Substrate)** 1. Pancreas; digests emulsified fats into fatty acids and glycerol
2. Gastric glands (stomach); digests proteins into peptones in acidic medium
3. Salivary glands (mouth); digests starch into maltose in slightly acidic/neutral medium
4. Exocrine Pancreas; digests proteins into peptides in alkaline medium

**Options:** A. A-2, B-3, C-4, D-1
B. A-3, B-2, C-4, D-1
C. A-2, B-4, C-3, D-1
D. A-4, B-3, C-2, D-1

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Pepsin:** Secreted as inactive pepsinogen by chief cells of stomach; activated by hydrochloric acid ($HCl$); optimum pH ~1.5–2.0; hydrolyzes proteins. - **Ptyalin (Salivary Amylase):** Secreted by salivary glands; begins carbohydrate digestion in mouth by converting starch into maltose. - **Trypsin:** Secreted as inactive trypsinogen by pancreas; activated by enterokinase in the small intestine; digests proteins in alkaline pH (~8.0). - **Lipase (Steapsin):** Secreted by pancreas and intestinal glands; breaks down fats emulsified by bile salts into fatty acids and glycerol. - **आयोग का Trap / Ratta Point:** Bile juice is secreted by the Liver and stored in the Gallbladder, but **Bile contains NO digestive enzymes**! It only emulsifies fats and neutralizes acidic chyme.

## Question 107

**Subject:** Uttar Pradesh Special | **Topic:** Major Industrial Belts of UP

Match List-I (Industrial City in UP) with List-II (Specialized Industry) and select the correct answer:

**List-I (City)** A. Moradabad
B. Firozabad
C. Khurja (Bulandshahr)
D. Modinagar (Ghaziabad)

**List-II (Specialization)** 1. Glass bangles and glassware
2. Brass handicraft and utensils (*Peetal Nagri*)
3. Synthetic textiles and chemicals
4. Ceramic tableware and glazed pottery

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Moradabad:** Famous globally as *Peetal Nagri* (Brass City) for exported brass handicrafts. - **Firozabad:** Known as *Suhag Nagari*; largest producer of glass bangles and decorative glass items in India. - **Khurja (Bulandshahr):** Famous across Asia as "Ceramic City" for bone china, ceramic tiles, and glazed pottery. - **Modinagar (Ghaziabad):** Industrial township established by Gujar Mal Modi, known for synthetic fibers, textiles, and vanaspati. - **आयोग का Trap / Ratta Point:** Churk and Dalla in Sonbhadra are famous for **Cement**; Bareilly is famous for **Matches (Safety Matches) and Cane Furniture**!

## Question 108

**Subject:** Environment & Ecology | **Topic:** Atmospheric Pollution & Acid Rain

Consider the following statements regarding Acid Rain: 1. Acid rain occurs when rainwater has a pH lower than 5.6. 2. The primary chemical precursors responsible for acid rain are Sulphur Dioxide ($SO_2$) and Oxides of Nitrogen ($NO_x$). 3. The yellowing and corrosion of the Taj Mahal's marble is known as "Stone Cancer", caused by acid precipitation reacting with calcium carbonate.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - Normal unpolluted rainwater is slightly acidic ($ ext{pH} □pprox 5.6$) due to dissolved atmospheric $CO_2$ forming weak carbonic acid. Any precipitation with $ ext{pH} < 5.6$ is classified as **Acid Rain**. - $SO_2$ and $NO_x$ emitted from coal thermal plants, refineries (like Mathura Refinery near Agra), and automobiles react with water vapor: $$SO_2 + 🡅rac{1}{2}O_2 + H_2O ightarrow H_2SO_4 ext{ (Sulphuric Acid)}$$ $$2NO_2 + 🡅rac{1}{2}O_2 + H_2O ightarrow 2HNO_3 ext{ (Nitric Acid)}$$ - Reaction with marble ($CaCO_3$): $$CaCO_3 + H_2SO_4 ightarrow CaSO_4 + H_2O + CO_2$$ Resulting calcium sulphate flakes away, corroding the monuments ("Marble Cancer"). - **आयोग का Trap / Ratta Point:** Acid rain produces sulphuric acid (~60–70%) and nitric acid (~30–40%). Hydrochloric acid is NOT a major component of typical acid rain.

## Question 109

**Subject:** Indian Economy | **Topic:** Banking Sector & Insolvency and Bankruptcy Code (IBC)

Consider the following statements regarding the Insolvency and Bankruptcy Code (IBC), 2016: 1. The National Company Law Tribunal (NCLT) is the adjudicating authority for insolvency resolution of corporate persons and companies. 2. The Debt Recovery Tribunal (DRT) is the adjudicating authority for insolvency resolution of individuals and partnership firms. 3. The Insolvency and Bankruptcy Board of India (IBBI) regulates insolvency professionals and information utilities.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - IBC 2016 created a unified legal framework for time-bound resolution of stressed assets: - **NCLT:** Adjudicating Authority for corporate debtors, LLPs, and companies. - **DRT:** Adjudicating Authority for individuals and partnership firms. - **IBBI:** Apex regulatory body set up under the Code (oversees IPAs, IPs, and Information Utilities like NeSL). - Corporate Insolvency Resolution Process (CIRP) must be completed within 180 days (extendable by 90 days, with an outer ceiling of 330 days). - **आयोग का Trap / Ratta Point:** Remember the jurisdictional boundary: Corporates go to NCLT; Individuals go to DRT!

## Question 110

**Subject:** Current Affairs | **Topic:** Sports & ICC Men's T20 World Cup 2024

Consider the following statements regarding the ICC Men's T20 World Cup 2024: 1. India won the championship by defeating South Africa by 7 runs in the final played at Kensington Oval, Bridgetown, Barbados. 2. Jasprit Bumrah was awarded the 'Player of the Tournament'. 3. The 2024 tournament was jointly co-hosted by the West Indies and the United States of America (USA).

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - India won its 2nd Men's T20 World Cup title (after 2007) on June 29, 2024, remaining unbeaten throughout the tournament. - Captained by Rohit Sharma; Player of the Match in Final was Virat Kohli (76 runs); Player of the Tournament was **Jasprit Bumrah** (15 wickets at an economy rate of 4.17). - It was the first ICC World Cup to feature 20 teams and the first hosted in the USA. - **आयोग का Trap / Ratta Point:** Both Rohit Sharma and Virat Kohli announced their retirement from T20 Internationals immediately after winning this final!

## Question 111

**Subject:** Ancient Indian History | **Topic:** Gupta Coinage & Epigraphic Titles

Match List-I (Gupta Ruler) with List-II (Coin Type / Imperial Title) and select the correct answer:

**List-I (Ruler)** A. Chandragupta I
B. Samudragupta
C. Chandragupta II
D. Kumaragupta I

**List-II (Coin / Epithet)** 1. Ashvamedha-type and Veena-playing type gold coins; *Kaviraja*
2. King-and-Queen (Kumaradevi) type gold coins; *Maharajadhiraja*
3. Silver coins introduced for the first time; *Vikramaditya* and *Sakari*
4. Kartikeya / Peacock-type gold coins; *Mahendraditya*

**Options:** A. A-2, B-1, C-3, D-4
B. A-1, B-2, C-3, D-4
C. A-2, B-3, C-1, D-4
D. A-4, B-1, C-3, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Chandragupta I:** First Gupta ruler to assume the title *Maharajadhiraja*; issued the famous joint Chandragupta I-Kumaradevi (Lichchhavi) matrimonial gold coins. - **Samudragupta:** Depicted playing the lute/veena on his coins; celebrated as *Kaviraja* (King of poets) and *Param-Bhagavata*; performed the Ashvamedha sacrifice. - **Chandragupta II:** Defeated the Western Kshatrapas (Sakas) in Malwa/Gujarat; assumed titles *Sakari* and *Vikramaditya*; was the FIRST Gupta king to issue **Silver coins** (*Rupaka*). - **Kumaragupta I:** Issued peacock-riding Kartikeya coins; assumed title *Mahendraditya*; founded the celebrated **Nalanda Mahavihara**. - **आयोग का Trap / Ratta Point:** Gold coins in Gupta era were called **Dinars**; silver coins were called **Rupakas**!

## Question 112

**Subject:** World Geography | **Topic:** Temperate Grasslands of the World

Match List-I (Temperate Grassland) with List-II (Country / Continent) and select the correct answer:

**List-I (Grassland)** A. Prairies
B. Pampas
C. Veld
D. Downs

**List-II (Region)** 1. Argentina and Uruguay (South America)
2. North America (USA and Canada)
3. Australia
4. South Africa

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Prairies:** North American interior (Mississippi-Missouri basin); "Granary of the World" (commercial wheat belt). - **Pampas:** Fertile South American plains in Argentina and Uruguay; famous for *Alfa-alfa* grass and cattle ranching (*Estancias*). - **Veld:** High plateau of South Africa; temperate pastoral grassland (Merino sheep). - **Downs:** Murray-Darling basin of Southeastern Australia; wheat and sheep farming. - **आयोग का Trap / Ratta Point:** Steppes = Eurasia (Russia/Ukraine); Puszta = Hungary; Canterbury = New Zealand. Tropical grasslands include Llanos (Venezuela), Campos (Brazil), and Savanna (Africa).

## Question 113

**Subject:** Indian Polity | **Topic:** Supreme Court Jurisdiction & Powers

Consider the following statements regarding the Supreme Court of India: 1. Under Article 131, the Supreme Court has exclusive Original Jurisdiction in disputes between the Government of India and one or more States. 2. Under Article 143, the President can seek the advisory opinion of the Supreme Court, and the opinion delivered by the court is legally binding on the President. 3. Under Article 142, the Supreme Court may pass such decree or make such order as is necessary for doing complete justice in any cause or matter.

Which of the statements given above is/are correct?

**Options:** A. 1 and 3 only
B. 1 and 2 only
C. 2 and 3 only
D. 1, 2 and 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Statement 1 is correct: Article 131 gives original and exclusive jurisdiction over federal disputes (Center vs State, Center & State vs State, State vs State). It does NOT extend to inter-state water disputes (Article 262) or ordinary commercial contracts. - Statement 2 is incorrect: Under Article 143 (Advisory Jurisdiction), the Supreme Court's advice is **purely advisory and NOT binding on the President**! (The President is free to accept or ignore the advice). - Statement 3 is correct: Article 142 empowers the apex court with plenary power to pass any order necessary for doing "complete justice" between the parties. - **आयोग का Trap / Ratta Point:** Under Article 143(1) (question of public importance), SC *may* decline to answer. But under Article 143(2) (pre-constitutional treaties), SC *must* give its opinion!

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## Question 114

**Subject:** Medieval Indian History | **Topic:** Maratha Administration (Ashtapradhan of Shivaji)

Match List-I (Ashtapradhan Officer) with List-II (Portfolio / Responsibility) and select the correct answer:

**List-I (Officer)** A. Peshwa (Mukhya Pradhan)
B. Amatya (Majumdar)
C. Sumant (Dabir)
D. Sachiv (Surunavis)

**List-II (Responsibility)** 1. Finance and Revenue Minister (Accountant General)
2. Foreign Affairs Minister
3. Prime Minister / General Administration
4. Royal Secretariat and Official Correspondence

**Options:** A. A-3, B-1, C-2, D-4
B. A-1, B-3, C-2, D-4
C. A-3, B-2, C-1, D-4
D. A-2, B-1, C-4, D-3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Shivaji was assisted by a council of eight ministers known as the **Ashtapradhan**: - **Peshwa:** Prime Minister looking after general administration. - **Amatya / Majumdar:** Finance minister in charge of public accounts. - **Sumant / Dabir:** Minister for Foreign Affairs. - **Sachiv / Shurunavis:** Handled royal correspondence and royal seals. - **Senapati / Sari-Naubat:** Commander-in-Chief of the army. - **Waqia-Navis / Mantri:** Home affairs, intelligence, and court chronicler. - **Nyayadhish:** Chief Justice. - **Pandit Rao:** Religious affairs and royal charities. - **आयोग का Trap / Ratta Point:** All ministers in the Ashtapradhan EXCEPT Nyayadhish and Pandit Rao were required to lead military expeditions in times of war!

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## Question 115

**Subject:** General Science (Physics) | **Topic:** Magnetic & Nuclear Constants

Match List-I (Physical Quantity) with List-II (SI / Scientific Unit) and select the correct answer:

**List-I (Quantity)** A. Magnetic Flux
B. Magnetic Flux Density (Magnetic Field)
C. Nuclear Radius / Size
D. Luminous Intensity

**List-II (Unit)** 1. Tesla (T)
2. Weber (Wb)
3. Candela (cd)
4. Fermi (Femtometre, $10^{-15} ext{ m}$)

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Magnetic Flux ($\Phi$):** Measured in **Weber (Wb)**. $1 ext{ Wb} = 1 ext{ V} \cdot ext{s} = 1 ext{ T} \cdot ext{m}^2$. - **Magnetic Field ($B$):** Measured in **Tesla (T)** in SI system (in CGS, it is Gauss; $1 ext{ Tesla} = 10^4 ext{ Gauss}$). - **Nuclear Radius:** Measured in **Fermi (fm)**, where $1 ext{ Fermi} = 10^{-15} ext{ metres}$. - **Luminous Intensity:** One of the 7 base SI units, measured in **Candela (cd)**. - **आयोग का Trap / Ratta Point:** Do not confuse Magnetic Flux (Weber) with Magnetic Field / Flux Density (Tesla). Magnetic Flux = Field $ imes$ Area!

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## Question 116

**Subject:** Uttar Pradesh Special | **Topic:** National Research Institutes in UP

Match List-I (National Research Institute) with List-II (Location in Uttar Pradesh) and select the correct answer:

**List-I (Institute)** A. Central Drug Research Institute (CDRI)
B. Indian Institute of Vegetable Research (IIVR)
C. Indian Institute of Pulses Research (IIPR)
D. National Sugar Institute (NSI)

**List-II (City in UP)** 1. Varanasi
2. Kanpur
3. Lucknow
4. Kanpur

*Note: Match carefully between IIPR and NSI in Kanpur.*

**Options:** A. A-3, B-1, C-2, D-4 (or A-3, B-1, C-4, D-2)
B. A-1, B-3, C-2, D-4
C. A-3, B-2, C-1, D-4
D. A-4, B-1, C-2, D-3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **CDRI (Central Drug Research Institute):** Located at Jankipuram, **Lucknow** (CSIR institute; also NBRI, CIMAP, and IITR are in Lucknow). - **IIVR (Indian Institute of Vegetable Research):** Located at **Varanasi** (ICAR institute). - **IIPR (Indian Institute of Pulses Research):** Located at Kalyanpur, **Kanpur**. - **NSI (National Sugar Institute):** Located at Kalyanpur, **Kanpur** (Note: Indian Institute of Sugarcane Research - IISR is in Lucknow!). - **आयोग का Trap / Ratta Point:** Massive trap: - National Sugar Institute (NSI) = **Kanpur** - Indian Institute of Sugarcane Research (IISR) = **Lucknow**!

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## Question 117

**Subject:** Environment & Ecology | **Topic:** Types of Smog (Photochemical vs Classical)

Consider the following statements regarding atmospheric smog: 1. Photochemical smog (Los Angeles smog) is an oxidizing smog formed in warm, sunny, and dry climates by the reaction of sunlight on nitrogen oxides and hydrocarbons. 2. Ozone ($O_3$) and Peroxyacetyl Nitrate (PAN) are key secondary pollutants present in photochemical smog. 3. Classical smog (London smog) is a reducing smog that occurs in cool, humid climates and contains a high mixture of smoke, fog, and sulphur dioxide ($SO_2$).

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Photochemical Smog (Los Angeles type):** Formed during sunny summer afternoons; auto exhaust produces $NO_2$ and unburnt hydrocarbons; UV rays photolyze $NO_2$ releasing free oxygen atoms that generate ground-level Ozone ($O_3$), PAN (Peroxyacetyl Nitrate), and formaldehyde; chemically **Oxidizing**. - **Classical Smog (London type):** Formed in winter mornings with temperature inversion; coal smoke + fog + $SO_2$; chemically **Reducing**. - **आयोग का Trap / Ratta Point:** Photochemical smog is OXIDIZING (due to high ozone/peroxides); Classical smog is REDUCING (due to sulphur dioxide)!

## Question 118

**Subject:** Modern Indian History | **Topic:** Reforms of Lord William Bentinck (1828–1835)

Consider the following reforms undertaken during the Governor-Generalship of Lord William Bentinck: 1. Abolition of the practice of Sati through Regulation XVII of December 1829. 2. Suppression of the menace of Thuggee organized under Captain William Sleeman. 3. Passing of the English Education Act of 1835 based on Lord Macaulay's Minute.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Abolition of Sati:** Passed on December 4, 1829, as Regulation XVII of the Bengal Code, declaring Sati illegal and punishable as culpable homicide (extended to Madras and Bombay Presidencies in 1830), actively supported by Raja Ram Mohan Roy. - **Thuggee Suppression:** Bentinck appointed Captain William Henry Sleeman in 1830; hundreds of hereditary robbers/stranglers (Thugs) were arrested and suppressed. - **English Education:** Act of 1835 endorsed Macaulay's famous Minute of February 1835, making English the official medium of higher instruction and government transactions. - **आयोग का Trap / Ratta Point:** Bentinck was the last Governor-General of Bengal and became the **FIRST Governor-General of India** under the Charter Act of 1833!

## Question 119

**Subject:** Indian Economy | **Topic:** Concepts of Inflation

Match List-I (Economic Concept) with List-II (Definition / Impact) and select the correct answer:

**List-I (Concept)** A. Core Inflation
B. Headline Inflation
C. Stagflation
D. Demand-Pull Inflation

**List-II (Definition)** 1. Measures total inflation in an economy including food and fuel items
2. Inflation that excludes volatile price components (food and energy commodities)
3. Situation of stagnant economic growth and high unemployment accompanied by high inflation
4. Price rise driven by aggregate demand exceeding aggregate supply in an economy

**Options:** A. A-2, B-1, C-3, D-4
B. A-1, B-2, C-3, D-4
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Core Inflation:** Headline inflation minus Food and Fuel; tracks underlying non-volatile inflation trend unaffected by short-term supply shocks. - **Headline Inflation:** Comprehensive measure of all goods and services in the consumer basket (CPI-Combined in India). - **Stagflation:** Portmanteau of stagnation and inflation; high unemployment, slowing GDP growth, and rising inflation (defies the classic short-run Phillips curve). - **Demand-Pull Inflation:** "Too much money chasing too few goods"; occurs when aggregate demand outpaces productive capacity. - **आयोग का Trap / Ratta Point:** Cost-Push Inflation is caused by supply-side shocks, like sudden spikes in international crude oil prices or agricultural crop failures!

## Question 120

**Subject:** Current Affairs | **Topic:** International Affairs & SCO Summit 2024

Consider the following statements regarding the Shanghai Cooperation Organisation (SCO): 1. The 24th Meeting of the Council of Heads of State of the SCO was held in Astana, Kazakhstan, in July 2024. 2. Belarus was formally admitted as the 10th full member state of the SCO at the Astana Summit. 3. India hosted the in-person SCO Council of Heads of Government (CHG) summit in October 2024.

Which of the statements given above is/are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 only
D. 1, 2 and 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Statements 1 and 2 are correct: The SCO summit in Astana, Kazakhstan (July 3–4, 2024) saw **Belarus** officially inducted as the 10th member state (Iran was admitted as the 9th member in 2023 under India's chairmanship). - Statement 3 is incorrect: The SCO Council of Heads of Government (CHG) meeting in October 2024 was hosted by **Islamabad, Pakistan** (EAM S. Jaishankar represented India, marking the first visit by an Indian Foreign Minister to Pakistan in nine years). - **आयोग का Trap / Ratta Point:** Current 10 members of SCO: China, Russia, Kazakhstan, Kyrgyzstan, Tajikistan, Uzbekistan, India (2017), Pakistan (2017), Iran (2023), Belarus (2024).

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## Question 121

**Subject:** Geography of India | **Topic:** Peninsular Mountain Sequences

Arrange the following mountain and hill ranges of Peninsular India in sequential order from NORTH to SOUTH: 1. Satpura Range 2. Ajanta Range 3. Nilgiri Hills 4. Cardamom Hills

Select the correct answer from the codes given below:

**Options:** A. 1 — 2 — 3 — 4
B. 2 — 1 — 3 — 4
C. 1 — 3 — 2 — 4
D. 1 — 2 — 4 — 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Sequential alignment from North to South: 1. **Satpura Range (MP/Maharashtra border):** Between Narmada and Tapi rivers (~21° to 22° N). 2. **Ajanta Range (Maharashtra):** South of Tapi river in Marathwada (~19° to 20° N). 3. **Nilgiri Hills (TN/Kerala/Karnataka junction):** Meeting point of Western and Eastern Ghats (~11° N). 4. **Cardamom Hills (Kerala/Tamil Nadu):** Southernmost hill range of mainland India (~9° N). - **आयोग का Trap / Ratta Point:** Anaimalai Hills lie between Nilgiri Hills (separated by Palghat gap) and Cardamom Hills. Sequence North to South: **Nilgiri -> Anaimalai -> Cardamom Hills**!

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## Question 122

**Subject:** Indian Polity | **Topic:** Anti-Defection Law & 91st Amendment Act

Consider the following statements regarding the 91st Constitutional Amendment Act, 2003: 1. It limited the total number of ministers, including the Prime Minister, in the Union Council of Ministers to not exceed 15% of the total strength of the Lok Sabha. 2. It mandated that the total number of ministers, including the Chief Minister, in a State shall not exceed 15% of the total strength of the Legislative Assembly, subject to a minimum of 12 ministers. 3. It deleted the provision in the Tenth Schedule that previously protected defecting legislators if one-third of a parliamentary party split.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - Enacted in 2003 to curb jumbo cabinets and horse-trading: - **15% Ceiling:** Union Council of Ministers cannot exceed 15% of Lok Sabha membership (Article 75(1A)). - **State Ceiling:** State Council of Ministers cannot exceed 15% of Legislative Assembly strength, but cannot be less than 12 ministers (Article 164(1A)). - **Anti-Defection:** Removed the "split" defense (former paragraph 3 of Tenth Schedule); now only a **merger** of not less than two-thirds ($\frac{2}{3}$) of the members of a legislative party protects them from disqualification. - **आयोग का Trap / Ratta Point:** In Delhi (NCT), under Article 239AA, the Council of Ministers cannot exceed **10%** of the Legislative Assembly (maximum 7 ministers including CM)!

## Question 123

**Subject:** Ancient Indian History | **Topic:** Jain Tirthankaras and Their Cognizance Symbols

Match List-I (Jain Tirthankara) with List-II (Lanchhana / Cognizance Symbol) and select the correct answer:

**List-I (Tirthankara)** A. Rishabhanatha (Adinatha)
B. Parshvanatha
C. Mahavira
D. Neminatha (Arishtanemi)

**List-II (Symbol)** 1. Serpent hood (Snake)
2. Bull (*Vrishabha*)
3. Conch shell (*Shankha*)
4. Lion (*Simha*)

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Rishabhanatha (1st Tirthankara):** Bull (*Vrishabha*); born at Ayodhya. - **Parshvanatha (23rd Tirthankara):** Serpent (*Sarpa*); prince of Varanasi; propounded four vows: Ahimsa, Satya, Asteya, Aparigraha. - **Mahavira (24th Tirthankara):** Lion (*Simha*); born at Kundagrama (Vaishali); added the fifth vow: Brahmacharya (celibacy). - **Neminatha (22nd Tirthankara):** Conch shell (*Shankha*); considered contemporary/cousin of Krishna. - **Mallinatha (19th Tirthankara):** Water pot (*Kalasha*); regarded as female by Svetambara sect. - **आयोग का Trap / Ratta Point:** Sambhavnatha (3rd Tirthankara) has the **Horse** as his symbol; Ajitanatha (2nd) has the **Elephant**!

## Question 124

**Subject:** General Science (Physics) | **Topic:** Anomalous Expansion of Water & Latent Heat

Consider the following statements regarding the thermal properties of matter: 1. Water exhibits anomalous expansion between 0°C and 4°C, reaching its maximum density at 4°C. 2. Because water expands as it cools from 4°C to 0°C, ice floats on water and aquatic life survives in frozen lakes. 3. Steam at 100°C produces more severe burns than boiling water at 100°C because steam possesses extra latent heat of vaporization (approx. 540 cal/g).

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Anomalous Expansion:** Unlike other liquids which contract uniformly upon cooling, water contracts until 4°C, where it reaches **maximum density (1 g/cm$^3$ or 1000 kg/m$^3$)** and minimum volume. Below 4°C, it expands. - **Aquatic Survival:** In winter, surface water cools to 4°C and sinks; when surface reaches 0°C, it freezes into ice. Since ice is a poor conductor of heat and less dense than water, bottom water remains liquid at 4°C, allowing aquatic animals to survive. - **Latent Heat:** Steam contains latent heat of vaporization ($2.26 imes 10^6 ext{ J/kg}$ or $□pprox 540 ext{ cal/g}$); when condensing on skin, it releases this enormous latent heat at constant temperature before cooling as water. - **आयोग का Trap / Ratta Point:** At 4°C, water has MAXIMUM density and MINIMUM volume!

## Question 125

**Subject:** Uttar Pradesh Special | **Topic:** District Boundaries Touching Neighboring States

Consider the following statements regarding district borders of Uttar Pradesh: 1. Saharanpur district shares borders with three states: Uttarakhand, Himachal Pradesh, and Haryana. 2. Sonbhadra district shares borders with four states: Madhya Pradesh, Chhattisgarh, Jharkhand, and Bihar. 3. Exactly seven districts of Uttar Pradesh share an international border with Nepal.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Saharanpur:** Tri-state border (Uttarakhand, Himachal Pradesh, Haryana). - **Sonbhadra:** Quad-state border (Madhya Pradesh, Chhattisgarh, Jharkhand, Bihar). - **7 Nepal Border Districts:** From West to East: 1. Pilibhit 2. Lakhimpur Kheri 3. Bahraich 4. Shravasti 5. Balrampur 6. Siddharthnagar 7. Maharajganj (Total Indo-Nepal border length in UP $□pprox 579 ext{ km}$). - **आयोग का Trap / Ratta Point:** Mnemonic for 7 UP districts touching Nepal: **P-L-B-S-B-S-M** (Pilibhit, Lakhimpur, Bahraich, Shravasti, Balrampur, Siddharthnagar, Maharajganj).

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## Question 126

**Subject:** Environment & Ecology | **Topic:** Mangrove Ecosystems in India

Consider the following statements regarding mangroves in India: 1. Mangrove plants exhibit specialized respiratory roots called 'pneumatophores' that grow vertically upwards to obtain atmospheric oxygen. 2. Mangroves exhibit 'viviparous germination', where seeds germinate while still attached to the parent tree before falling into saline mud. 3. West Bengal has the largest mangrove forest cover in India, followed by Gujarat and Andaman & Nicobar Islands.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - Mangroves survive in anaerobic saline intertidal mudflats via: - **Pneumatophores (blind roots):** Grow negative geotropically with lenticels for gas exchange. - **Stilt roots:** Provide mechanical anchorage against ocean waves. - **Vivipary:** Seed germinates into a seedling while still attached to mother plant. - **State Rankings (ISFR):** 1. **West Bengal** (~42.45% of India's total mangrove cover in Sundarbans). 2. **Gujarat** (~23.66% in Gulf of Kutch and Khambhat). 3. **Andaman & Nicobar Islands** (~12.39%). - **आयोग का Trap / Ratta Point:** Second largest mangrove forest in India is **Bhitarkanika** (Odisha, located in the delta of Brahmani and Baitarani rivers).

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## Question 127

**Subject:** Modern Indian History | **Topic:** Women Presidents of Indian National Congress

Match List-I (Woman President of INC) with List-II (Year & Session) and select the correct answer:

**List-I (President)** A. Annie Besant
B. Sarojini Naidu
C. Nellie Sengupta

**List-II (Session & Year)** 1. 1925 Kanpur Session; First Indian woman President of Congress
2. 1917 Calcutta Session; First woman President of Congress
3. 1933 Calcutta Session; Third woman President of Congress

**Options:** A. A-2, B-1, C-3
B. A-1, B-2, C-3
C. A-2, B-3, C-1
D. A-3, B-1, C-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Annie Besant (1917 Calcutta):** First EVER woman to preside over the Indian National Congress. - **Sarojini Naidu (1925 Kanpur):** First **INDIAN** woman to preside over the Congress; celebrated as *Nightingale of India* (*Bharat Kokila*). - **Nellie Sengupta (1933 Calcutta):** English-born woman married to J.M. Sengupta; third woman to preside over Congress in pre-independence India. - **आयोग का Trap / Ratta Point:** Gap of exactly 8 years: **1917 (Annie Besant) + 8 = 1925 (Sarojini Naidu) + 8 = 1933 (Nellie Sengupta)**!

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## Question 128

**Subject:** Indian Polity & Governance | **Topic:** Structure and Functions of NITI Aayog

Consider the following statements regarding the NITI Aayog: 1. It was established by an executive resolution of the Union Cabinet on January 1, 2015, replacing the Planning Commission. 2. The Prime Minister of India serves as the ex-officio Chairperson of NITI Aayog. 3. The Governing Council of NITI Aayog comprises the Prime Minister, Chief Ministers of all States, and Lieutenant Governors of Union Territories.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Establishment:** NITI (National Institution for Transforming India) Aayog was constituted on 1 January 2015 as a non-statutory, non-constitutional think-tank emphasizing "Cooperative Federalism" and "Bottom-up approach". - **Leadership:** Chairperson = Prime Minister. Vice-Chairperson is appointed by the PM (enjoys rank of Cabinet Minister). - **Governing Council:** Chief Ministers of all States, CMs of UTs with legislatures (Delhi, Puducherry, J&K), and Lt. Governors of other UTs. - **आयोग का Trap / Ratta Point:** NITI Aayog does NOT possess the power to allocate financial funds to states; that power rests with the Ministry of Finance and the Finance Commission!

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## Question 129

**Subject:** Current Affairs | **Topic:** Padma Vibhushan Awards 2024

Who among the following was conferred the Padma Vibhushan (India's second-highest civilian honor) posthumously in 2024 for Public Affairs / Social Work?

**Options:** A. Bindeshwar Pathak
B. M. Venkaiah Naidu
C. Konidela Chiranjeevi
D. Vyjayanthimala Bali

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - In 2024, the President approved 5 Padma Vibhushan awards: 1. **Bindeshwar Pathak (Posthumous):** Founder of Sulabh International; sanitation pioneer and social worker. 2. **Ms. Vyjayanthimala Bali:** Veteran classical dancer and cinema actress (Art). 3. **Konidela Chiranjeevi:** Renowned Telugu actor (Art). 4. **M. Venkaiah Naidu:** Former Vice-President of India (Public Affairs). 5. **Ms. Padma Subrahmanyam:** Celebrated Bharatanatyam dancer and scholar (Art). - **आयोग का Trap / Ratta Point:** Remember: Bindeshwar Pathak was the only posthumous recipient among the 5 Padma Vibhushan awardees in 2024.

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## Question 130

**Subject:** Medieval Indian History | **Topic:** Vijayanagara Empire Dynasties Chronology

Arrange the four ruling dynasties of the Vijayanagara Empire in chronological order: 1. Saluva Dynasty 2. Sangama Dynasty 3. Aravidu Dynasty 4. Tuluva Dynasty

Select the correct answer from the codes given below:

**Options:** A. 2 — 1 — 4 — 3
B. 1 — 2 — 4 — 3
C. 2 — 4 — 1 — 3
D. 2 — 1 — 3 — 4

**Answer:** A

**Explanation:** - **Core Fact & Solution:** 1. **Sangama Dynasty (1336–1485):** Founded by brothers Harihara I and Bukka I; prominent rulers: Deva Raya I and Deva Raya II. 2. **Saluva Dynasty (1485–1505):** Founded by Saluva Narasimha (First Usurpation). 3. **Tuluva Dynasty (1505–1570):** Founded by Vira Narasimha (Second Usurpation); reached its zenith under **Krishnadeva Raya (1509–1529)**; suffered disaster at the Battle of Talikota (1565). 4. **Aravidu Dynasty (1570–1646):** Founded by Tirumala Raya with capital at Penukonda and Chandragiri. - **आयोग का Trap / Ratta Point:** Mnemonic: **S-S-T-A** (Sangama -> Saluva -> Tuluva -> Aravidu). Battle of Talikota (Rakshasi-Tangadi) was fought on 23 January 1565 during the reign of Sadashiva Raya (Tuluva dynasty).

## Question 131

**Subject:** World Geography | **Topic:** Ocean Trenches & Deepest Points

Match List-I (Ocean Trench / Deep) with List-II (Ocean Location) and select the correct answer:

**List-I (Trench)** A. Mariana Trench (Challenger Deep)
B. Puerto Rico Trench (Milwaukee Deep)
C. Java / Sunda Trench
D. Molloy Deep

**List-II (Ocean)** 1. Atlantic Ocean
2. Pacific Ocean
3. Arctic Ocean
4. Indian Ocean

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Mariana Trench (Challenger Deep):** Deepest oceanic point on Earth (~10,994 m); Western Pacific Ocean near Guam. - **Puerto Rico Trench (Milwaukee Deep):** Deepest point in the Atlantic Ocean (~8,376 m). - **Java / Sunda Trench:** Deepest point in the Indian Ocean (~7,450 m). - **Molloy Deep:** Deepest point in the Arctic Ocean (~5,550 m, in Fram Strait). - **आयोग का Trap / Ratta Point:** Pacific Ocean contains the maximum number of oceanic trenches because it is ringed by subduction zones ("Pacific Ring of Fire").

## Question 132

**Subject:** Indian Polity | **Topic:** Emergency Provisions (Part XVIII)

Consider the following statements regarding Emergency Provisions: 1. Under Article 352, a Proclamation of National Emergency must be approved by both Houses of Parliament by a special majority within one month. 2. The 44th Constitutional Amendment Act substituted the phrase "internal disturbance" with "armed rebellion" in Article 352. 3. During a National Emergency, the right to protection in respect of conviction for offences (Article 20) and right to life and personal liberty (Article 21) cannot be suspended.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Approval & Majority:** Article 352 proclamation must be approved within 1 month by a special majority (majority of total membership and $\ge ↟rac{2}{3}$ of members present and voting) in each House; valid for 6 months. - **44th CAA 1978 Safeguards:** - Replaced ambiguous "internal disturbance" with "armed rebellion". - Mandated written advice of the Union Cabinet (only place "Cabinet" is mentioned in Constitution: Article 352(3)). - **Articles 20 and 21 can NEVER be suspended** even under emergency (Article 359). - **आयोग का Trap / Ratta Point:** Article 19 is automatically suspended ONLY when National Emergency is declared on grounds of "War" or "External Aggression" (External Emergency), NOT on grounds of "Armed Rebellion" (Internal Emergency)!

## Question 133

**Subject:** General Science (Biology) | **Topic:** Cell Organelles and Functions

Match List-I (Cell Organelle) with List-II (Popular Epithet / Biochemical Function) and select the correct answer:

**List-I (Organelle)** A. Mitochondria
B. Lysosome
C. Ribosome
D. Golgi Apparatus

**List-II (Epithet / Function)** 1. "Suicidal Bag" of the cell; contains hydrolytic enzymes
2. "Powerhouse of the cell"; site of cellular respiration & ATP synthesis
3. Packaging and secretion center; formation of acrosome in sperm
4. "Protein factory" of the cell; site of translation

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Mitochondria:** Double-membraned organelle; site of Krebs cycle and oxidative phosphorylation generating ATP; possesses its own circular DNA and 70S ribosomes (semi-autonomous). - **Lysosome:** Single-membraned vesicle filled with acid hydrolases (optimum pH ~5.0) that digest foreign material and worn-out cell parts ("Suicidal Bags"). - **Ribosome:** Non-membrane bound ribonucleoprotein particle; synthesizes polypeptide chains from mRNA templates ("Protein Factories"). - **Golgi Apparatus:** Discovered by Camillo Golgi; modifies, sorts, and packages macromolecules for secretion; forms lysosomes and the sperm acrosome. - **आयोग का Trap / Ratta Point:** Both Mitochondria and Chloroplasts are **semi-autonomous organelles** having their own DNA and 70S ribosomes (like bacteria, supporting endosymbiotic theory)!

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## Question 134

**Subject:** Uttar Pradesh Special | **Topic:** Agricultural Output & Superlatives

In which of the following agricultural commodities does Uttar Pradesh hold the FIRST rank in production in India? 1. Wheat 2. Sugarcane 3. Potato 4. Total Foodgrains

Select the correct answer from the codes given below:

**Options:** A. 1, 2 and 3 only
B. 1, 3 and 4 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - Uttar Pradesh holds the **FIRST rank** in India in: - **Wheat** (produces over 32% of India's wheat) - **Sugarcane** (produces over 45% of India's sugarcane) - **Potato** (produces over 30% of India's potato) - **Total Foodgrains** (~18–19% of India's total output) - **Milk production** (competing closely with Rajasthan for top rank) - In **Rice production**, UP ranks **SECOND** in India (West Bengal ranks first). - **आयोग का Trap / Ratta Point:** UP ranks 1st in Wheat, Sugarcane, Potato, Mango, Amla, and Total Foodgrains; but ranks 2nd in Rice!

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## Question 135

**Subject:** Environment & Ecology | **Topic:** CITES Appendices Classification

Consider the following statements regarding the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES): 1. Appendix I includes species threatened with extinction, and international commercial trade in these specimens is generally prohibited. 2. Appendix II includes species that are not necessarily threatened with extinction now, but may become so unless trade is strictly regulated. 3. CITES Secretariat is administered by the United Nations Environment Programme (UNEP) and is located in Geneva, Switzerland.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **CITES (Washington Convention, 1973):** Legally binding on Parties; protects over 38,000 species. - **Appendix I:** Highest protection (Tiger, Asiatic Lion, Great Indian Bustard, Red Panda); commercial trade banned. - **Appendix II:** Trade permitted only with export permits certifying that trade will not be detrimental to survival in the wild. - **Appendix III:** Contains species protected in at least one country that has asked CITES parties for assistance. - **Headquarters:** Geneva, Switzerland; administered by UNEP. - **आयोग का Trap / Ratta Point:** CITES regulates ONLY international commercial trade of wild specimens; it does not replace domestic wildlife management laws!

## Question 136

**Subject:** Modern Indian History | **Topic:** Swaraj Party (1923)

Consider the following statements regarding the Swaraj Party: 1. It was founded in January 1923 at Allahabad by Chittaranjan Das (President) and Motilal Nehru (Secretary). 2. The core objective of the Swarajists was to enter the legislative councils and "end or mend" them from within. 3. In August 1925, Vithalbhai J. Patel was elected as the first Indian Speaker (President) of the Central Legislative Assembly.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - Formed following the Gaya session (December 1922) where C.R. Das's proposal for council entry was defeated by "No-Changers" (Rajendra Prasad, Vallabhbhai Patel, C. Rajagopalachari). - In January 1923, Das and Motilal Nehru announced the formation of the **Congress-Khilafat Swarajya Party** at Allahabad. - In the November 1923 elections, Swarajists won 42 out of 101 elected seats in the Central Legislative Assembly. - Major constitutional victory: On August 24, 1925, **Vithalbhai Patel** was elected President (Speaker) of the Central Legislative Assembly! - **आयोग का Trap / Ratta Point:** Pro-Changers = C.R. Das, Motilal Nehru, Hakim Ajmal Khan; No-Changers = Sardar Patel, Dr. Rajendra Prasad, C. Rajagopalachari, M.A. Ansari.

## Question 137

**Subject:** Indian Economy | **Topic:** Balance of Payments (BoP) Structure

Match List-I (BoP Component) with List-II (Sub-head / Item) and select the correct answer:

**List-I (Component)** A. Current Account
B. Capital Account
C. Foreign Direct Investment (FDI)
D. Foreign Portfolio Investment (FPI)

**List-II (Item / Characteristic)** 1. External Commercial Borrowings (ECB), NRI deposits, and foreign loans
2. Trade in goods (merchandise), trade in services, remittances, and gifts
3. Passive investment in stocks/bonds (<10% equity) without managerial control
4. Non-debt creating capital flow with lasting interest ( $\ge 10\%$ equity) and active management

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Current Account:** Encompasses Visible trade (exports/imports of goods), Invisible trade (software services, tourism, shipping), and Unilateral transfers (remittances from NRIs, gifts, grants). - **Capital Account:** Encompasses FDI, FPI, External Commercial Borrowings (ECB), foreign aid, NRI bank deposits, and sovereign loans (affects asset/liability position). - **FDI vs FPI (Mayaram Committee Definition):** Foreign investment of **10% or more** in the listed equity of an Indian company is classified as FDI (stable, long-term). Investment of **less than 10%** is classified as FPI (volatile, "hot money"). - **आयोग का Trap / Ratta Point:** India usually runs a **Current Account Deficit (CAD)** due to heavy crude oil/gold imports, but compensates for it with a **Capital Account Surplus**!

## Question 138

**Subject:** Current Affairs | **Topic:** Sports & 45th Chess Olympiad 2024

Consider the following statements regarding the 45th Chess Olympiad held in Budapest, Hungary in September 2024: 1. India scripted history by winning Gold medals in both the Open and Women's team sections. 2. Gukesh D and Arjun Erigaisi won individual Gold medals on their respective boards in the Open section. 3. This was the first time India won the Gold medal in the Women's section at the Chess Olympiad.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - At the 45th FIDE Chess Olympiad in Budapest (September 2024), India achieved an unprecedented **Double Gold**: - **Open Team:** Gukesh D, Praggnanandhaa R, Arjun Erigaisi, Vidit Gujrathi, Pentala Harikrishna. - **Women's Team:** Harika Dronavalli, Vaishali Rameshbabu, Divya Deshmukh, Vantika Agrawal, Tania Sachdev. - Board 1 individual gold went to D. Gukesh (performance rating 3056); Board 3 gold went to Arjun Erigaisi. - India became only the 3rd country in history (after Soviet Union and China) to win both team golds in the same Olympiad. - **आयोग का Trap / Ratta Point:** D. Gukesh also became the youngest player ever to challenge for the World Chess Championship!

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## Question 139

**Subject:** Ancient Indian History | **Topic:** Vedic Literature & Upanishadic Maxims

Match List-I (Upanishad) with List-II (Famous Philosophical Maxim / Dialogue) and select the correct answer:

**List-I (Upanishad)** A. Mundaka Upanishad
B. Kathopanishad
C. Chandogya Upanishad
D. Brihadaranyaka Upanishad

**List-II (Maxim / Dialogue)** 1. "Satyameva Jayate" (Truth alone triumphs)
2. Dialogue between young seeker Nachiketa and Lord Yama on the nature of death and soul
3. "Tamaso Ma Jyotirgamaya" and dialogue between sage Yajnavalkya and Maitreyi
4. First mention of Krishna as the son of Devaki and dialogue of Uddalaka Aruni with Svetaketu

**Options:** A. A-1, B-2, C-4, D-3
B. A-2, B-1, C-4, D-3
C. A-1, B-4, C-2, D-3
D. A-3, B-2, C-1, D-4

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Mundaka Upanishad:** Source of India's national motto "*Satyameva Jayate*"; also compares rituals and sacrifices to "leaky boats" (*plava hyete adridha yajnarupah*). - **Kathopanishad:** Famous philosophical dialogue between the young boy Nachiketa and Yama (God of Death) regarding the immortality of Atman. - **Chandogya Upanishad:** Mentions Krishna, son of Devaki, as a disciple of sage Ghora Angirasa; contains the famous phrase "*Tat Tvam Asi*" (That Thou Art); mentions first three ashramas. - **Brihadaranyaka Upanishad:** Largest Upanishad; source of "*Tamaso Ma Jyotirgamaya*"; records the celebrated debate between sage Yajnavalkya and philosopher Gargi Vachaknavi at King Janaka's court. - **आयोग का Trap / Ratta Point:** All four Ashramas (Brahmacharya, Grihastha, Vanaprastha, Sannyasa) are mentioned together for the first time in the **Jabala Upanishad**!

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## Question 140

**Subject:** World Geography | **Topic:** Major Rivers of the World & Drainage

Match List-I (River) with List-II (Outflow Sea / Ocean & Feature) and select the correct answer:

**List-I (River)** A. Nile River
B. Amazon River
C. Yangtze River
D. Mississippi-Missouri

**List-II (Outflow & Feature)** 1. Atlantic Ocean; world's largest river by water discharge and drainage basin
2. Mediterranean Sea; world's longest river (~6,650 km)
3. Gulf of Mexico; forms a prominent "Bird's Foot Delta"
4. East China Sea; longest river in Asia (~6,300 km)

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Nile:** Longest river in the world; flows northward through 11 African nations into the Mediterranean Sea; formed by the confluence of White Nile (originates Lake Victoria) and Blue Nile (Lake Tana, Ethiopia) at Khartoum (Sudan). - **Amazon:** Largest river by volume and basin area; originates in Peruvian Andes; empties into the Atlantic Ocean; carries more water than the next 7 largest rivers combined. - **Yangtze (Chang Jiang):** Longest river in Asia and third longest in the world; Three Gorges Dam is situated on it; empties into East China Sea near Shanghai. - **Mississippi-Missouri:** Longest river system in North America; famous for its distinctive **Bird's foot delta** in the Gulf of Mexico. - **आयोग का Trap / Ratta Point:** River Congo (Zaire) crosses the Equator TWICE! River Limpopo (South Africa) crosses the Tropic of Capricorn TWICE! River Mahi (India) crosses the Tropic of Cancer TWICE!

## Question 141

**Subject:** Indian Polity | **Topic:** Landmark Constitutional Amendment Acts

Match List-I (Constitutional Amendment Act) with List-II (Core Subject) and select the correct answer:

**List-I (Amendment)** A. 42nd Amendment Act, 1976
B. 44th Amendment Act, 1978
C. 86th Amendment Act, 2002
D. 103rd Amendment Act, 2019

**List-II (Subject)** 1. Right to Education (Article 21A) inserted as a Fundamental Right
2. "Mini-Constitution"; added Fundamental Duties, 3 words to Preamble, and 4 DPSPs
3. 10% Reservation for Economically Weaker Sections (EWS) in education and public employment
4. Right to Property removed from Fundamental Rights; safeguards against Emergency abuse

**Options:** A. A-2, B-4, C-1, D-3
B. A-4, B-2, C-1, D-3
C. A-2, B-1, C-4, D-3
D. A-3, B-4, C-1, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **42nd CAA 1976:** Known as "Mini-Constitution"; based on Swaran Singh Committee; added "Socialist, Secular, Integrity" to Preamble; Part IV-A (Fundamental Duties); Articles 39A, 43A, 48A. - **44th CAA 1978 (Janata Govt):** Removed Right to Property from Part III (made a legal right under Article 300A); restored Lok Sabha term to 5 years; inserted safeguards for Emergency under Article 352/359. - **86th CAA 2002:** Inserted Article 21A (free and compulsory education for ages 6–14); substituted Article 45 in DPSPs; added Article 51A(k) (11th fundamental duty). - **103rd CAA 2019:** Amended Articles 15(6) and 16(6) to grant up to 10% reservation to Economically Weaker Sections (EWS). - **आयोग का Trap / Ratta Point:** 101st CAA = GST; 102nd CAA = Constitutional status to National Commission for Backward Classes (NCBC - Article 338B); 104th CAA = Extended SC/ST seats in LS/Assemblies, ended Anglo-Indian nomination; 106th CAA = Nari Shakti Vandan Adhiniyam (33% women's reservation)!

## Question 142

**Subject:** General Science (Chemistry) | **Topic:** Alloys and Their Compositions

Match List-I (Alloy) with List-II (Constituent Metals) and select the correct answer:

**List-I (Alloy)** A. Brass
B. Bronze
C. German Silver
D. Solder (Soldering Wire)

**List-II (Constituent Metals)** 1. Copper ($Cu$) and Tin ($Sn$)
2. Copper ($Cu$) and Zinc ($Zn$)
3. Lead ($Pb$) and Tin ($Sn$)
4. Copper ($Cu$), Zinc ($Zn$), and Nickel ($Ni$)

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Brass:** Alloy of Copper (~70%) and Zinc (~30%); used in utensils, cartridges, and musical instruments. - **Bronze:** Alloy of Copper (~88%) and Tin (~12%); used in statues, medals, and coins. - **German Silver:** Alloy of Copper (~50%), Zinc (~30%), and Nickel (~20%). **It contains NO SILVER (0% Ag)!** Named solely for its silvery-white luster. - **Solder:** Alloy of Lead (~50%) and Tin (~50%); has a low melting point, used for joining electrical wires. - **Stainless Steel:** Iron ($Fe$) + Chromium ($Cr \ge 10.5\%$) + Nickel ($Ni$) + Carbon ($C$). - **आयोग का Trap / Ratta Point:** Recurring UPPSC trap: "Which of the following elements is NOT present in German Silver?" Answer: **Silver ($Ag$)**!

## Question 143

**Subject:** Uttar Pradesh Special | **Topic:** Traditional Theatrical & Puppet Arts of UP

Consider the following statements regarding folk theatre and puppetry in Uttar Pradesh: 1. 'Nautanki' is a prominent folk-theatre of Uttar Pradesh having two distinct stylistic schools: the Hathras style and the Kanpur style. 2. 'Gulabo-Sitabo' is a traditional glove-puppetry art form originating from the Awadh region (Lucknow) of Uttar Pradesh. 3. In Nautanki, the musical dialogue is performed accompanied by instruments like Nagara (kettledrum), Dholak, and Harmonium.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Nautanki:** Highly popular musical folk theatre form across Northern India, especially UP. - *Hathras Style:* Traditional, emphasizes classical singing and sophisticated metered verse (*Doha, Chaubola, Chhand*). - *Kanpur Style:* More dramatic, contemporary, action-oriented, with colloquial language. - **Gulabo-Sitabo:** Rare traditional glove puppet theatre unique to Lucknow; revolves around the comical bickering between two co-wives (Gulabo, the vibrant mistress, and Sitabo, the weary wife), reflecting Awadhi social satire. - **आयोग का Trap / Ratta Point:** String puppetry of Rajasthan is *Kathputli*; Shadow puppetry of Andhra is *Tholu Bommalata*; Glove puppetry of UP is *Gulabo-Sitabo*!

## Question 144

**Subject:** Environment & Ecology | **Topic:** Stratospheric Ozone Layer Depletion

Consider the following statements regarding the Ozone layer: 1. The protective ozone layer is concentrated primarily in the lower Stratosphere between 15 km and 30 km above the Earth's surface. 2. Ozone thickness in the atmosphere is measured in Dobson Units (DU). 3. Ground-level (tropospheric) ozone is a beneficial gas, while stratospheric ozone is a harmful air pollutant.

Which of the statements given above is/are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 only
D. 1, 2 and 3

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - Statements 1 and 2 are correct: Stratospheric ozone (~90% of atmospheric ozone) absorbs lethal solar ultraviolet radiation (UV-B and UV-C). Normal column thickness is ~300 Dobson Units (1 DU = 0.01 mm thickness at STP). An "ozone hole" is defined when concentration drops below 220 DU. - Statement 3 is incorrect: The situation is the EXACT REVERSE! **Stratospheric ozone is "Good Ozone"** (shields Earth from UV radiation). **Tropospheric (ground-level) ozone is "Bad Ozone"** (a toxic secondary pollutant and greenhouse gas causing respiratory distress and crop damage). - **आयोग का Trap / Ratta Point:** "Good Ozone" = Stratosphere; "Bad Ozone" = Troposphere! World Ozone Day is celebrated annually on **16th September** (commemorating the signing of Montreal Protocol in 1987).

## Question 145

**Subject:** Modern Indian History | **Topic:** Administration and Commissions of Lord Curzon (1899–1905)

Match List-I (Commission / Reform of Lord Curzon) with List-II (Chairman / Mandate) and select the correct answer:

**List-I (Curzon Reform)** A. Police Commission (1902)
B. Universities Commission (1902)

C. Irrigation Commission (1901)
D. Ancient Monuments Preservation Act (1904)

**List-II (Chairman / Mandate)** 1. Sir Thomas Raleigh; resulted in Indian Universities Act of 1904
2. Sir Andrew Fraser; recommended creation of CID (Criminal Investigation Department)
3. Established Archaeological Survey of India (ASI) on a permanent basis
4. Sir Colin Scott-Moncrieff; planned extensive canal systems in Punjab

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **Police Commission (1902):** Headed by Sir Andrew Fraser; reported that police administration was corrupt and inefficient; created Central CID and Provincial CID. - **Universities Commission (1902):** Headed by Sir Thomas Raleigh (included Syed Hussain Bilgrami and Justice Gurudas Banerjee); led to the **Indian Universities Act of 1904**, bringing universities under tight governmental control. - **Irrigation Commission (1901):** Headed by Sir Colin Scott-Moncrieff; led to massive canal expansion in the Punjab colony. - **Ancient Monuments Preservation Act (1904):** Curzon allocated funds to repair historical monuments, making the Archaeological Survey of India (ASI) a permanent government department under John Marshall. - **आयोग का Trap / Ratta Point:** Curzon partitioned Bengal on **October 16, 1905**; Bengal was reunited in **1911** by Lord Hardinge II at the Delhi Durbar!

## Question 146

**Subject:** Indian Economy | **Topic:** Multidimensional Poverty Index (MPI)

Consider the following statements regarding the National Multidimensional Poverty Index (MPI) published by NITI Aayog: 1. India's National MPI uses the Alkire-Foster methodology, retaining the 3 dimensions of Global MPI: Health, Education, and Standard of Living. 2. The National MPI measures poverty across 12 indicators (adding 'Maternal Health' and 'Bank Accounts' to the 10 global indicators). 3. As per NITI Aayog's MPI reports, Uttar Pradesh registered the largest decline in the number of multidimensionally poor individuals among all states.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Dimensions & Weights:** 3 equally weighted dimensions (Health $\frac{1}{3}$, Education $\frac{1}{3}$, Standard of Living $\frac{1}{3}$). - **12 Indicators:** Health (Nutrition, Child & Adolescent Mortality, Maternal Health); Education (Years of Schooling, School Attendance); Standard of Living (Cooking Fuel, Sanitation, Drinking Water, Electricity, Housing, Assets, Bank Accounts). - **State Progress:** As per NITI Aayog's discussion paper (2024), ~24.82 crore people escaped multidimensional poverty in India between 2013–14 and 2022–23. **Uttar Pradesh recorded the largest escape from poverty (approx. 5.94 crore people)**, followed by Bihar and Madhya Pradesh. - **आयोग का Trap / Ratta Point:** Highest multidimensionally poor population by percentage remains Bihar, but the **largest absolute reduction** occurred in Uttar Pradesh!

## Question 147

**Subject:** Current Affairs | **Topic:** Climate Summits (COP28 & COP29)

Consider the following statements regarding the UN Climate Change Conferences: 1. COP28 was held in Dubai, United Arab Emirates (UAE) in December 2023, where the 'Loss and Damage Fund' was formally operationalized on day one. 2. COP28 concluded with the historic "UAE Consensus", calling upon countries to transition away from fossil fuels in energy systems. 3. COP29 in November 2024 was hosted by Baku, Azerbaijan, focusing on the New Collective Quantified Goal (NCQG) on climate finance.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **COP28 (Dubai, UAE 2023):** Presided by Sultan Al Jaber. Breakthrough operationalization of the Loss and Damage Fund (hosted initially by the World Bank). The "Global Stocktake" agreed on transitioning away from fossil fuels to achieve net zero by 2050. - **COP29 (Baku, Azerbaijan, Nov 2024):** Dubbed the "Finance COP"; central objective was replacing the outdated \$100 billion/year climate finance target with the **New Collective Quantified Goal (NCQG)**. - **COP30 (2025):** Scheduled to be held in Belém (Amazon region), Brazil. - **आयोग का Trap / Ratta Point:** COP sequence: COP27 = Sharm el-Sheikh (Egypt) -> COP28 = Dubai (UAE) -> COP29 = Baku (Azerbaijan) -> COP30 = Belém (Brazil).

## Question 148

**Subject:** Geography of India | **Topic:** National Waterways of India

Match List-I (National Waterway) with List-II (River Route / Stretch) and select the correct answer:

**List-I (Waterway)** A. National Waterway 1 (NW-1)
B. National Waterway 2 (NW-2)
C. National Waterway 3 (NW-3)
D. National Waterway 4 (NW-4)

**List-II (Route / Stretch)** 1. Brahmaputra River; Sadiya to Dhubri (Assam)
2. Ganga-Bhagirathi-Hooghly river system; Prayagraj (Allahabad) to Haldia
3. Krishna and Godavari rivers with Buckingham Canal; Kakinada to Puducherry
4. West Coast Canal; Kottapuram to Kollam (Kerala)

**Options:** A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

**Answer:** A

**Explanation:** - **Core Fact & Solution:** - **NW-1:** Longest national waterway in India (1,620 km); spans Ganga-Bhagirathi-Hooghly from **Prayagraj (UP) to Haldia (WB)**; passes through UP, Bihar, Jharkhand, and WB. - **NW-2:** Brahmaputra River from **Sadiya to Dhubri** in Assam (891 km). - **NW-3:** West Coast Canal from **Kottapuram to Kollam** along with Champakara and Udyogmandal canals in Kerala (205 km). - **NW-4:** Connects Kakinada to Puducherry along Godavari and Krishna rivers and Buckingham canal (1,078 km). - **आयोग का Trap / Ratta Point:** Inland Waterways Authority of India (IWAI) was statutory body created in 1986; headquarters located at **Noida (Uttar Pradesh)**!

## Question 149

**Subject:** Indian Polity | **Topic:** State Legislative Councils (Article 169)

Consider the following statements regarding State Legislative Councils (Vidhan Parishad): 1. Under Article 169, Parliament can abolish or create a Legislative Council in a State if the Legislative Assembly of that State passes a resolution by a special majority. 2. The total membership of the Legislative Council shall not exceed one-third of the total membership of the Legislative Assembly, but cannot be less than 40 members. 3. At present, six states in India possess a bicameral legislature: Uttar Pradesh, Bihar, Maharashtra, Karnataka, Andhra Pradesh, and Telangana.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Creation/Abolition (Article 169):** State Assembly must pass resolution by a special majority ($\ge ↟rac{2}{3}$ present & voting and majority of total membership). Parliament then passes an ordinary law by simple majority (not deemed an amendment under Article 368). - **Composition (Article 171):** Maximum strength = $↟rac{1}{3}$ of Legislative Assembly; Minimum strength = 40. UP Legislative Council has the highest strength with **100 seats**. - **6 Bicameral States: Uttar Pradesh, Bihar, Maharashtra, Karnataka, Andhra Pradesh, and Telangana** (Jammu & Kashmir had a council until its bifurcation in 2019). - **आयोग का Trap / Ratta Point:** Mnemonic for 6 states with Legislative Councils: **KUMBAT** (Karnataka, UP, Maharashtra, Bihar, Andhra Pradesh, Telangana)!

## Question 150

**Subject:** Uttar Pradesh Special | **Topic:** Irrigation Sources & Major Canals in UP

Consider the following statements regarding irrigation in Uttar Pradesh: 1. Tubewells (Nalkoop) constitute the largest source of irrigation in Uttar Pradesh, accounting for over 70% of the total net irrigated area. 2. Canals form the second largest source of irrigation in the state, accounting for approximately 15% of irrigated area. 3. The Sharda Canal is the longest and largest canal network in Uttar Pradesh, originating from Banbasa on the Sharda river.

Which of the statements given above are correct?

**Options:** A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

**Answer:** D

**Explanation:** - **Core Fact & Solution:** - All three statements are correct: - **Irrigation Sources Share in UP:** 1. **Tubewells (Nalkoop):** ~73.4% (Highest in UP; first tubewell dug in Meerut in 1930). 2. **Canals (Nahar):** ~15.2%. 3. **Wells (Kuan):** ~8.6%. 4. **Tanks, Lakes & Ponds:** ~1.2%. - **Canal Networks:** - **Sharda Canal:** Largest and longest canal system in UP; built between 1920–1928; originates at Banbasa (Champawat border) on the Sharda (Kali) river. - **Upper Ganga Canal:** Oldest major modern canal in northern India; built between 1842–1854 by Sir Proby Cautley from Haridwar. - **Eastern Yamuna Canal:** Oldest canal in UP (originating 1830 from Faizabad, Saharanpur). - **आयोग का Trap / Ratta Point:** Tubewells account for nearly three-fourths of all irrigation in UP; canals account for only about one-seventh!